

S.A ENGINEERING COLLEGE, CHENNAI – 77
(An Autonomous Institution Affiliated to Anna University)
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
REGULATION-2020
CHOICE BASED CREDIT SYSTEM
CURRICULUM AND SYLLABUS

VISION

To conceive our department as Centre of Academic Excellence by catering quality education with ethical standards

MISSION

M1: To create a conducive atmosphere to achieve active professionalism by fortifying academic proficiency with ethical standards.

M2: To enhance the confidence level to develop sustainable solution to upgrade the society forever.

M3: To empower the students with prerequisite professional skills for enhancing employability and entrepreneurship, continuing education and research.

PROGRAM EDUCATIONAL OBJECTIVES (PEOs):

PEO1: Graduates shall have professional competency in the field of Computer Science and Engineering for pursuing higher education, research or as entrepreneurs.

PEO2: Graduates shall work in a business environment with ethical standards, leadership qualities and communication necessary for engineering principles.

PEO3: Graduates shall adapt to emerging technologies and respond to the challenges of the environment and society forever.

PROGRAM OUTCOMES (POs):

Engineering Graduates will be able to:

1. **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.
2. **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental

considerations.

4. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5. **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
6. **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9. **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10. **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11. **Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
12. **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs):

PSO1: To analyze, design and develop computing solutions by applying foundational concepts of Computer Science and Engineering.

PSO 2: To apply software engineering principles and practices for developing quality software for scientific and business applications.

Mapping of POs/PSOs to PEOs

Contribution

1: Reasonable

2: Significant

3: Strong

MAPPING OF PROGRAM EDUCATIONAL OBJECTIVES WITH PROGRAM OUTCOMES

A broad relation between the program objective and outcomes is given in the following table:

Program Educational Objectives	Program Outcomes											
	1	2	3	4	5	6	7	8	9	10	11	12
1	3	3	3	3	3	2	2	3	3	3	2	3
2	2	2	2	3	2	2	1	3	3	3	3	2
3	3	3	3	3	3	1	3	1	2	3	2	3

MAPPING OF PROGRAM SPECIFIC OBJECTIVES WITH PROGRAM OUTCOMES

A broad relation between the program specific objective and outcomes is given in the following table:

Program Specific Objectives	Program Outcomes											
	1	2	3	4	5	6	7	8	9	10	11	12
1	3	3	3	2	3	3	3	3	2	2	3	3
2	3	3	3	2	3	2	2	2	2	2	2	3

MAPPING OF COURSE OUTCOMES WITH PROGRAM OUTCOMES

A broad relation between the Course Outcomes and Program Outcomes is given in the following table

YEAR-I	SEMESTER - I	Course Title	PROGRAMME OUTCOME(PO)											
			1	2	3	4	5	6	7	8	9	10	11	12
		Technical English	-	-	-	-	-	√	√	-	√	√	√	√
		Calculus and its applications	√	√	√	√	-	-	-	-	-	-	-	-
		Applied Physics	√	√	-	-	-	-	-	-	-	-	-	-
		Engineering Chemistry	√	√	√	-	-	-	-	-	-	-	-	-
		Problem Solving and Python Programming	√	√	√	√	√	-	-	-	-	-	√	√
		Engineering Graphics	√	√	√	√	√	√	√	√	√	√	√	√
		Tamil Marabu/ Heritage of Tamils												
		Physics and Chemistry Laboratory	√	√	√	√	-	-	-	-	-	-	-	-
		Problem Solving and Python Programming Laboratory	√	√	√	√	-	-	-	-	-	-	-	√
		Indian Constitution												
	SEMESTER - II	English for Communication	-	-	-	-	√	√	-	-	√	√	-	√
		Complex Variables and Transforms	√	√	√	-	-	-	-	-	-	-	-	-
		Material Science	√	√	-	-	-	-	-	√	-	-	-	-
		Basic Electrical, Electronics and Measurement Engineering	√	√	√	-	-	-	-	-	-	-	-	-
		Programming in C	√	√	√	√	-	-	-	-	-	-	√	√
		Fundamentals of Computing	√	√	√	-	-	-	-	-	-	-	-	√
		Programming in C Laboratory	√	√	√	√	-	-	-	-	-	-	-	
		Engineering Practice Laboratory	√	√	√	√	√	√	-	√	√	√	-	√

YEAR-II		Fundamentals of Computing Laboratory	√	√	√	√	√	-	-	√	√	√	√	√
		Environmental Science and Engineering	√	√	√	-	-	-	√	√	√	√	-	√
	PROGRAMME OUTCOME(PO)													
	SEMESTER - III	COURSE TITLE	1	2	3	4	5	6	7	8	9	10	11	12
		Discrete Mathematics	√	√	√	√	-	-	-	-	-	-	-	-
		Object Oriented Programming	√	√	√	√	√	-	-	-	√	-	-	√
		Data Structures	√	√	√	-	-	-	-	-	-	-	-	√
		Digital Principles and System Design	√	√	√	√	√	√	√	√	-	-	√	√
		Software Engineering	√	√	√	-	√	-	-	√	√	√	√	√
		Object Oriented Programming Laboratory	√	√	√	-	√	-	-	-	√	√	-	√
		Data Structures Laboratory	√	√	√	-	-	-	-	-	√	√	-	√
		Interpersonal Skills Laboratory	√	√	√	√	√	√	√	√	√	√	√	√
	SEMESTER - IV	Probability and Statistics	√	√	√	√	-	-	-	-	-	-	-	√
		Computer Architecture	√	√	√	-	-	-	-	-	-	-	-	√
		Database Management Systems	√	√	√	-	-	-	-	-	-	-	-	√
		Design and Analysis of Algorithms	√	√	√									√
		Universal Human Values	√	-	-	-	-	-	-	-	-	-	-	√
		Object Oriented Analysis and Design	√	√	√	√	√	-	-	-	√	√	-	√
		Database Management Systems Laboratory	√	√	√	-	√	-	-	-	√	√	√	√
		Design and Analysis of Algorithms Laboratory	√	√	√	-	-	-	-	-	√	√	-	√
		Employability and Soft Skills Laboratory	√	√	√	√	√	√	√	√	√	√	√	√

YEAR - III	SEMESTER - V	COURSE TITLE	PROGRAMME OUTCOME(PO)											
			1	2	3	4	5	6	7	8	9	10	11	12
			Algebra and Number Theory	√	√	√	-	-	-	-	-	-	-	-
			Computer Networks	√	√	√	-	-	-	-	-	-	-	√
			Operating Systems	√	√	√	-	-	-	-	-	-	-	√
			Theory of Computation	√	√	√	-	-	-	-	-	-	-	√
			Embedded System	√	√	√	√	√	-	-	-	√	√	√
			Open Elective I											
			Networks Laboratory	√	√	√	-	-	-	-	√	√	-	√
			Operating Systems Laboratory	√	√	√	-	-	-	-	-	-	-	-
	SEMESTER - VI	Internet Programming	√	√	√	-	√	-	√	-	√	√	√	√
		Compiler Design	√	√	√	-	√	-	-	-	-	-	-	√
		Artificial Intelligence	√	√	√	√	-	-	-	-	-	√	√	√
		Cryptography and Network Security	√	√	√	√	√	-	-	-	√	√	-	√
		Professional Elective I												
		Internet Programming Laboratory	√	√	√	√	√	-	√	-	√	√	√	√
		Professional Readiness for Innovation, Employability and Entrepreneurship	√	√	√	√	√	-	√	-	√	√	√	√
		Security Laboratory	√	√	√	√	√	-	-	-	-	-	-	√
		Professional Communication Laboratory	-	-	-	-	√	√	√	√	√	√	√	√

YEAR - IV	SEMESTER - VII	COURSE TITLE	PROGRAMME OUTCOME(PO)											
			1	2	3	4	5	6	7	8	9	10	11	12
			Distributed Systems and cloud computing	√	√	√	-	-	√	√	√	√	-	√
			Machine Learning	√	√	√		√	-	-	-	-	-	-
			Principles of Management	√		√	-	-	√	√	√	√	√	√
			Open Elective II											
			Professional Elective II											
			Professional Readiness for Innovation, Employability and Entrepreneurship	√	√	√	√	√	√	√	√	√	√	√
			Cloud Computing Laboratory	√	√	√	√	√	-	√	-	√	√	√

SEMESTER - VIII	Summer internship	√	√	√	√	√	√	√	√	√	√	√	√	√
	Machine Learning Laboratory	√	√	√	√	√	-	-	-	-	√	-	√	√
	Professional Elective III													
	Professional Elective IV													
	Project Work	√	√	√	√	√	√	√	√	√	√	√	√	√

PROFESSIONAL ELECTIVES

SEM	COURSE TITLE	PROGRAMME OUTCOME(PO)											
		1	2	3	4	5	6	7	8	9	10	11	12
VI	Cyber Security	√	√	√	√	√	√	√	√	√	√	√	√
	Software Testing	√	√	√	√	-	-	√	-	-	-	-	-
	Agile Methodologies	√	√	√								√	
	Graph Theory and Applications	√	√	√									
	Data warehousing and Data mining	√	√	√									
VII	Service Oriented Architecture	√	√	√								√	
	Total Quality Management	√	√	√									
	Multi-core Architectures and Programming	√	√	√									
	Human Computer Interaction	√	√	√	√	√	√	√	√	√	√	√	√
	C# and .Net Programming	√	√	√	√	√	√	-	-	-	-	-	√
	Wireless Adhoc and Sensor Networks	√	√	√									
	Advanced Topics on Databases	√	√	√									
	Full Stack Web Development	√	√	√	-	√	√	√	-	√	√	√	√
	Human Rights	√	√	√									
	Blockchain Technologies	√	√	√	√	√	√	√	√	-	-	-	√
VIII	Neural Networks and Deep Learning	√	√	√	√	√	√			√	√	√	√
	Professional Ethics in Engineering						√	√	√	√	√		√
	Big Data Analytics	√	√	√		√				√	√		
	Principles of Programming Languages	√	√	√									√
	Security and Privacy in Cloud	√	√	√	√	√				√	√	√	√
	Game Development	√	√	√	√	√	√						
	Multimedia and Animation	√	√	√	√	√	√			√	√	√	√
	UI and UX Design	√	√	√	√	-	-	-	-	-	-	-	-
VIII	Green Computing	√	√	√	√	√	√	√	-	√	√	√	√
	Mobile and Pervasive Computing	√	√	√									
	Software Defined Networks	√	√	√									
	Digital marketing	√	√	√	-	√	-	-	-	√	-	-	√
	Knowledge Engineering	√	√	√									√
	Ethical Hacking	√	√	√	√	√	√		√				√

SEMESTER - I									
SL.NO	SUBJECT CODE	SUBJECT	CATEGORY	CONTACT PERIODS	L	T	P	C	
THEORY									
1.	HS1101	Technical English	HS	3	3	0	0	3	
2.	MA1101	Calculus and its applications	BS	4	3	1	0	4	
3.	PH1101	Applied Physics	BS	3	3	0	0	3	
4.	CY1101	Engineering Chemistry	BS	3	3	0	0	3	
5.	CS1101	Problem Solving and Python Programming	ES	3	3	0	0	3	
6.	ME1101	Engineering Graphics	ES	4	2	0	2	3	
7.	TA1101	Tamil Marathi	HSMC	1	1	0	0	1	
PRACTICALS									
8.	BS1101	Physics and Chemistry Laboratory	BS	4	0	0	4	2	
9.	CS1102	Problem Solving and Python Programming Laboratory	ES	4	0	0	4	2	
MANDATORY COURSE									
10.	CI1101	Indian Constitution	MC	2	2	0	0	0	
TOTAL				31	20	1	10	24	

SEMESTER - II									
SL.NO	SUBJECT CODE	SUBJECT	CATEGORY	CONTACT PERIODS	L	T	P	C	
THEORY									
1.	HS1201	English for Communication	HS	3	3	0	0	3	
2.	MA1201	Complex Variables and Transforms	BS	4	3	1	0	4	
3.	PH1201	Materials Science	BS	3	3	0	0	3	
4.	EE1202	Basic Electrical, Electronics and Measurement Engineering	ES	3	3	0	0	3	
5.	CS1201	Programming in C	ES	3	3	0	0	3	
6.	CS1202	Fundamentals of Computing	PC	3	3	0	0	3	
7.	TA1201	Tamil Thozhil Nutpam	HSMC	1	1	0	0	1	
PRACTICALS									
8.	CS1203	Programming in C Laboratory	ES	4	0	0	4	2	
9.	GE1201	Engineering Practices Laboratory	ES	4	0	0	4	2	
10.	CS1204	Fundamentals of Computing Laboratory	PC	2	0	0	2	1	
MANDATORY COURSE									
11.	CY1201	Environmental Science and Engineering	MC	2	2	0	0	0	
TOTAL				32	21	1	10	25	

SEMESTER – III								
SL.NO	SUBJECT CODE	SUBJECT	CATEGORY	CONTACT PERIODS	L	T	P	C
THEORY								
1.	MA1303	Discrete Mathematics	BS	4	3	1	0	4
2.	IT1301	Object Oriented Programming	PC	3	3	0	0	3
3.	CS1301	Data Structures	PC	3	3	0	0	3
4.	EC1306	Digital Principles and System Design	ES	3	3	0	0	3
5.	CS1302	Software Engineering	PC	3	3	0	0	3
PRACTICALS								
6.	IT1302	Object Oriented Programming Laboratory	PC	4	0	0	4	2
7.	CS1303	Data Structures Laboratory	PC	4	0	0	4	2
8.	HS1301	Interpersonal Skills Laboratory	EEC	2	0	0	2	1
TOTAL				26	15	1	10	21

SEMESTER – IV								
SL.NO	SUBJECT CODE	SUBJECT	CATEGORY	CONTACT PERIODS	L	T	P	C
THEORY								
1.	MA1403	Probability and Statistics	BS	4	3	1	0	4
2.	CS1401	Computer Architecture	PC	3	3	0	0	3
3.	IT1401	Database Management Systems	PC	3	3	0	0	3
4.	CS1402	Design and Analysis of Algorithms	PC	3	3	0	0	3
5.	HV1401	Universal Human Values	ES	3	2	1	0	3
6.	CS1403	Object Oriented Analysis and Design	PC	5	3	0	2	4
PRACTICALS								
7.	IT1402	Database Management Systems Laboratory	PC	4	0	0	4	2
8.	CS1404	Design and Analysis of Algorithms Laboratory	PC	4	0	0	4	2
9.	HS1401	Employability And Soft Skills Laboratory	EEC	2	0	0	2	1
TOTAL				31	17	2	12	25

SEMESTER – V								
SL.NO	SUBJECT CODE	SUBJECT	CATEGORY	CONTACT PERIODS	L	T	P	C
THEORY								
1.	MA1501	Algebra and Number Theory	BS	4	3	1	0	4
2.	IT1501	Computer Networks	PC	3	3	0	0	3
3.	CS1501	Operating Systems	PC	3	3	0	0	3
4.	CS1502	Theory of Computation	PC	3	3	0	0	3
5.	EC1515	Embedded System	PC	3	3	0	0	3
6.		Open Elective-I	OE	3	3	0	0	3
PRACTICALS								
7.	IT1503	Networks Laboratory	PC	4	0	0	4	2
8.	CS1503	Operating Systems Laboratory	PC	4	0	0	4	2
TOTAL				27	18	1	8	23

SEMESTER –VI								
SL.NO	SUBJECT CODE	SUBJECT	CATEGORY	CONTACT PERIODS	L	T	P	C
THEORY								
1.	CS1601	Internet Programming	PC	3	3	0	0	3
2.	CS1602	Compiler Design	PC	5	3	0	2	4
3.	CS1603	Artificial Intelligence	PC	3	3	0	0	3
4.	CS1604	Cryptography and Network Security	PC	3	3	0	0	3
5.		Professional Elective-I	PE	3	3	0	0	3
PRACTICALS								
6.	CS1605	Internet Programming Laboratory	PC	4	0	0	4	2
7.	CS1606	Security Laboratory	PC	4	0	0	4	2
8.	ECS1601	Professional Readiness for Innovation, Employability and Entrepreneurship	EEC	6	0	0	6	0
9.	HS1601	Professional Communication Laboratory	EEC	2	0	0	2	1
TOTAL				33	15	0	18	21

SEMESTER – VII

SL.NO.	SUBJECT CODE	SUBJECT	CATEGORY	CONTACT PERIODS	L	T	P	C
THEORY								
1.	IT1701	Distributed Systems and Cloud Computing	PC	3	3	0	0	3
2.	CS1701	Machine Learning	PC	3	3	0	0	3
3.	MG1701	Principles of Management	HS	3	3	0	0	3
4.		Open Elective-II	OE	3	3	0	0	3
5.		Professional Elective-II	PE	3	3	0	0	3
PRACTICALS								
6.	CS1702	Machine Learning Laboratory	PC	4	0	0	4	2
7.	IT1702	Cloud Computing Laboratory	PC	4	0	0	4	2
8.	CS1703	Summer Internship	EEC	0	0	0	0	2
9.	ECS1601	Professional Readiness for Innovation, Employability and Entrepreneurship	EEC	0	0	0	0	3
TOTAL				23	15	0	8	24

SEMESTER – VIII

SL.NO.	SUBJECT CODE	SUBJECT	CATEGORY	CONTACT PERIODS	L	T	P	C
THEORY								
1.		Professional Elective-III	PE	3	3	0	0	3
2.		Professional Elective-IV	PE	3	3	0	0	3
PRACTICALS								
3.	CS1801	Project Work	EEC	20	0	0	20	10
TOTAL				26	6	0	20	16

TOTAL NO. OF CREDITS: 179

HUMANITIES AND SOCIAL SCIENCES (HS)

S.NO	SUBJECT CODE	SUBJECT	CATEGORY	CONTACT PERIODS	L	T	P	C
1.	HS1101	Technical English	HS	3	3	0	0	3
2.	HS1201	English for Communication	HS	3	3	0	0	3
3.	MG1701	Principles of Management	HS	3	3	0	0	3

HUMANITIES, SOCIAL SCIENCES AND MANAGEMENT COURSES (HSMC)

SL.NO.	SUBJECT CODE	SUBJECT	CATEGORY	CONTACT PERIODS	L	T	P	C
1.	TA1101	Tamilar Marabu	HSMC	1	1	0	0	1
2.	TA1201	Tamizhar Thozhil Nutpam	HSMC	1	1	0	0	1

BASIC SCIENCES (BS)

S.NO.	SUBJECT CODE	SUBJECT	CATEGORY	CONTACT PERIODS	L	T	P	C
1.	MA1101	Calculus and its applications	BS	4	3	1	0	4
2.	PH1101	Applied Physics	BS	3	3	0	0	3
3.	CY1101	Engineering Chemistry	BS	3	3	0	0	3
4.	BS1101	Physics and Chemistry Laboratory	BS	4	0	0	4	2
5.	MA1201	Complex Variables and Transforms	BS	4	3	1	0	4
6.	PH1201	Material Science	BS	3	3	0	0	3
7.	MA1303	Discrete Mathematics	BS	4	3	1	0	4
8.	MA1403	Probability and Statistics	BS	4	3	1	0	4
9.	MA1501	Algebra and Number Theory	BS	4	3	1	0	4

ENGINEERING SCIENCES (ES)

S.NO.	SUBJECT CODE	SUBJECT	CATEGORY	CONTACT PERIODS	L	T	P	C
1.	CS1201	Programming in C	ES	3	3	0	0	3
2.	CS1203	Programming in C Laboratory	ES	4	0	0	4	2
3.	CS1101	Problem Solving and Python Programming	ES	3	3	0	0	3
4.	ME1101	Engineering Graphics	ES	4	2	0	2	3
5.	EE1202	Basic Electrical, Electronics and Measurement Engineering	ES	3	3	0	0	3
6.	CS1102	Problem Solving and Python Programming Laboratory	ES	4	0	0	4	2
7.	GE1201	Engineering Practices Laboratory	ES	4	0	0	4	2
8.	EC1306	Digital Principles and System Design	ES	3	3	0	0	3
9.	HV1401	Universal Human Values	ES	3	2	1	0	3

EMPLOYABILITY ENHANCEMENT COURSES (EEC)								
Sl.NO	SUBJECT CODE	SUBJECT	CATEGORY	CONTACT PERIODS	L	T	P	C
1.	HS1301	Interpersonal Skills Laboratory	EEC	2	0	0	2	1
2.	HS1401	Employability And Soft Skills Laboratory	EEC	2	0	0	2	1
3.	HS1601	Professional Communication Laboratory	EEC	2	0	0	2	1
4.	CS1703	Summer Internship	EEC	0	0	0	0	2
5.	ECS1601	Professional Readiness for Innovation, Employability and Entrepreneurship	EEC	6	0	0	6	3
6.	CS1801	Project Work	EEC	20	0	0	20	10

PROFESSIONAL CORE (PC)								
Sl.NO	SUBJECT CODE	SUBJECT	CATEGORY	CONTACT PERIODS	L	T	P	C
1.	CS1202	Fundamentals of Computing	PC	3	3	0	0	3
2.	CS1204	Fundamentals of Computing Laboratory	PC	2	0	0	2	1
3.	IT1301	Object Oriented Programming	PC	3	3	0	0	3
4.	CS1301	Data Structures	PC	3	3	0	0	3
5.	CS1302	Software Engineering	PC	3	3	0	0	3
6.	IT1302	Object Oriented Programming Laboratory	PC	4	0	0	4	2
7.	CS1303	Data Structure Laboratory	PC	4	0	0	4	2
8.	CS1401	Computer Architecture	PC	3	3	0	0	3
9.	IT1401	Database Management Systems	PC	3	3	0	0	3
10.	CS1402	Design and Analysis of Algorithms	PC	3	3	0	0	3
11.	CS1403	Object Oriented Analysis and Design	PC	5	3	0	2	4
12.	IT1402	Database Management Systems Laboratory	PC	4	0	0	4	2
13.	CS1404	Design and Analysis of Algorithms Laboratory	PC	4	0	0	4	2
14.	IT1501	Computer Networks	PC	3	3	0	0	3
15.	CS1501	Operating Systems	PC	3	3	0	0	3
16.	CS1502	Theory of Computation	PC	3	3	0	0	3
17.	EC1515	Embedded System	PC	3	3	0	0	3
18.	IT1503	Networks Laboratory	PC	4	0	0	4	2
19.	CS1503	Operating Systems Laboratory	PC	4	0	0	4	2
20.	CS1601	Internet Programming	PC	3	3	0	0	3

21.	CS1602	Compiler Design	PC	5	3	0	2	4
22.	CS1603	Artificial Intelligence	PC	3	3	0	0	3
23.	CS1604	Cryptography and Network Security	PC	3	3	0	0	3
24.	CS1605	Internet Programming Laboratory	PC	4	0	0	4	2
25.	CS1606	Security Laboratory	PC	4	0	0	4	2
26.	IT1701	Distributed Systems and Cloud Computing	PC	3	3	0	0	3
27.	CS1701	Machine Learning	PC	3	3	0	0	3
28.	IT1702	Cloud Computing Laboratory	PC	4	0	0	4	2
29.	CS1702	Machine Learning Laboratory	PC	4	0	0	4	2

PROFESSIONAL ELECTIVES
ELECTIVE-I

Sl.NO	SUBJECT CODE	SUBJECT	CATEGORY	CONTACT PERIODS	L	T	P	C
THEORY:								
1.	CS1607	Cyber Security	PE	3	3	0	0	3
2.	CS1608	Software Testing	PE	3	3	0	0	3
3.	CS1609	Agile Methodologies	PE	3	3	0	0	3
4.	CS1610	Graph Theory and Applications	PE	3	3	0	0	3
5.	CS1611	Data warehousing and Data mining	PE	3	3	0	0	3

ELECTIVE-II

Sl.NO	SUBJECT CODE	SUBJECT	CATEGORY	CONTACT PERIODS	L	T	P	C
THEORY								
1.	CS1704	Service Oriented Architecture	PE	3	3	0	0	3
2.	CS1705	Total Quality Management	PE	3	3	0	0	3
3.	CS1706	Multi-core Architectures and Programming	PE	3	3	0	0	3
4.	CS1707	Human Computer Interaction	PE	3	3	0	0	3
5.	CS1708	C# and .Net Programming	PE	3	3	0	0	3
6.	CS1709	Wireless Adhoc and Sensor Networks	PE	3	3	0	0	3
7.	CS1710	Advanced Topics on Databases	PE	3	3	0	0	3
8.	CS1711	Full Stack Web Development	PE	3	3	0	0	3
9.	CS1712	Human Rights	PE	3	3	0	0	3
10.	CS1713	Blockchain Technologies	PE	3	3	0	0	3

ELECTIVE-III								
Sl.NO	SUBJECT CODE	SUBJECT	CATEGORY	CONTACT PERIODS	L	T	P	C
THEORY								
1.	GE1801	Professional Ethics in Engineering	PE	3	3	0	0	3
2.	CS1802	Neural Networks and Deep Learning	PE	3	3	0	0	3
3.	CS1803	Big Data Analytics	PE	3	3	0	0	3
4.	CS1804	Principles of Programming Languages	PE	3	3	0	0	3
5.	CS1805	Security and Privacy in Cloud	PE	3	3	0	0	3
6.	CS1806	Game Development	PE	3	3	0	0	3
7.	CS1807	Multimedia and Animation	PE	3	3	0	0	3

ELECTIVE-IV								
Sl.NO	SUBJECT CODE	SUBJECT	CATEGORY	CONTACT PERIODS	L	T	P	C
THEORY								
1.	CS1808	UI and UX Design	PE	3	3	0	0	3
2.	CS1809	Green Computing	PE	3	3	0	0	3
3.	CS1810	Mobile and Pervasive Computing	PE	3	3	0	0	3
4.	CS1811	Software Defined Networks	PE	3	3	0	0	3
5.	CS1812	Digital Marketing	PE	3	3	0	0	3
6.	CS1813	Knowledge Engineering	PE	3	3	0	0	3
7.	CS1814	Ethical Hacking	PE	3	3	0	0	3

OPEN ELECTIVE OFFERED TO OTHER DEPARTMENT
OPEN ELECTIVE I
SEMESTER V

Sl.NO	SUBJECT CODE	SUBJECT	CATEGORY	CONTACT PERIODS	L	T	P	C
THEORY								
1.	OCS501	Software Engineering	OE	3	3	0	0	3
2.	OCS502	Data Structures and Algorithms	OE	3	3	0	0	3
3.	OCS503	Telehealth Technology	OE	3	3	0	0	3

OPEN ELECTIVE II SEMESTER VII								
SL.NO	SUBJECT CODE	SUBJECT	CATEGORY	CONTACT PERIODS	L	T	P	C
THEORY								
	OCS701	Software Testing	OE	3	3	0	0	3
	OCS702	Web Design and management	OE	3	3	0	0	3

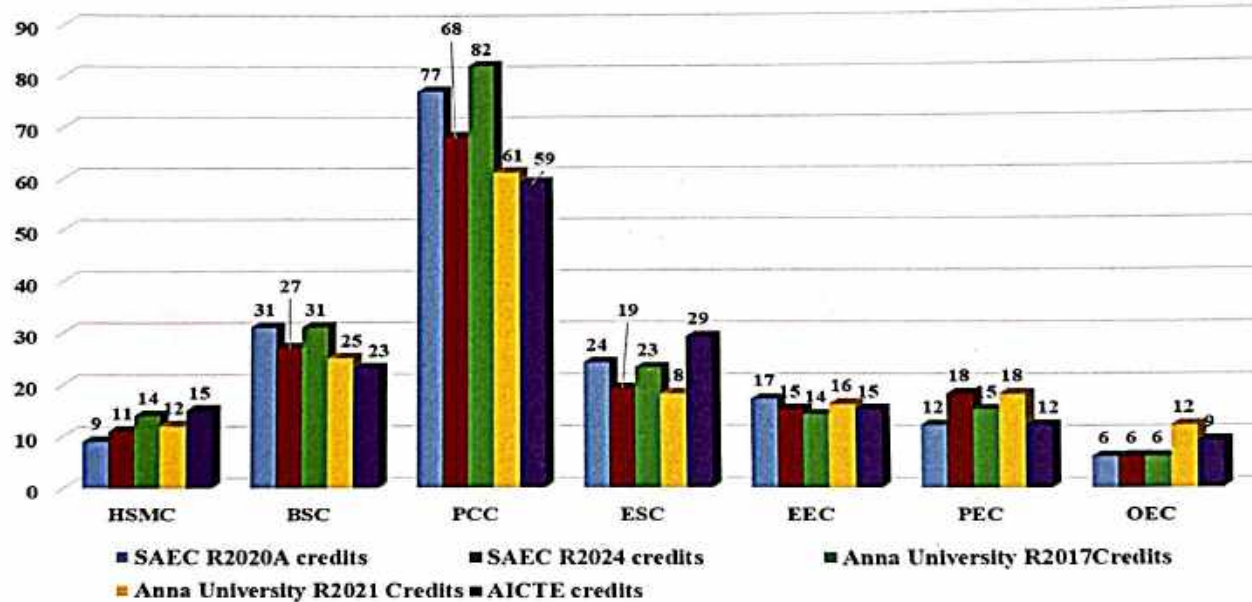
SUMMARY

Sl.No.	Subject Area	Credits per semester								Credits Total	Percentage %
		I	II	III	IV	V	VI	VII	VIII		
1.	HS	3	3					3		9	5.03
2.	BS	12	7	4	4	4				31	17.32
3.	ES	8	10	3	3					24	13.41
4.	PC		4	13	17	16	17	10		77	43.02
5.	PE						3	3	6	12	6.70
6.	OE					3		3		6	3.35
7.	EEC			1	1		1	5	10	18	10.06
8.	MC									0	0.00
9.	HS MC	1	1							2	1.12
TOTAL		24	25	21	25	23	21	24	16	179	

Alignment with AICTE/ University Guidelines

S.no	Category	SAEC R2020A credits	SAEC R2024 credits	Anna University R2017Credits	Anna University R2021 Credits	AICTE credits
1	HSMC	9	11	14	12	15
2	BSC	31	27	31	25	23
3	PCC	77	68	82	61	59
4	ESC	24	19	23	18	29
5	EEC	17	15	14	16	15
6	PEC	12	18	15	18	12
7	OEC	6	6	6	12	9
TOTAL		178	165	185	162	162

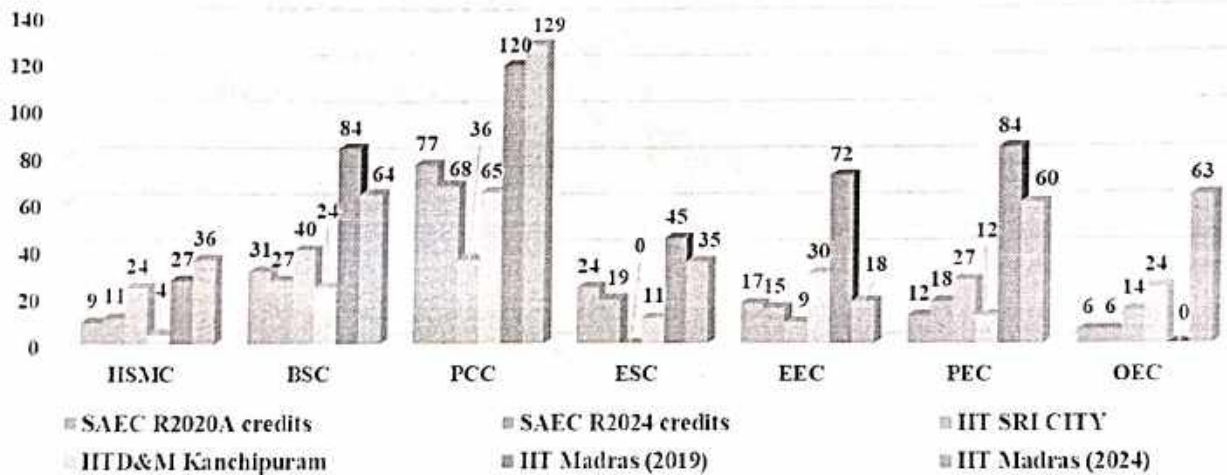
Alignment with AICTE/ University Guidelines



Alignment with IIT

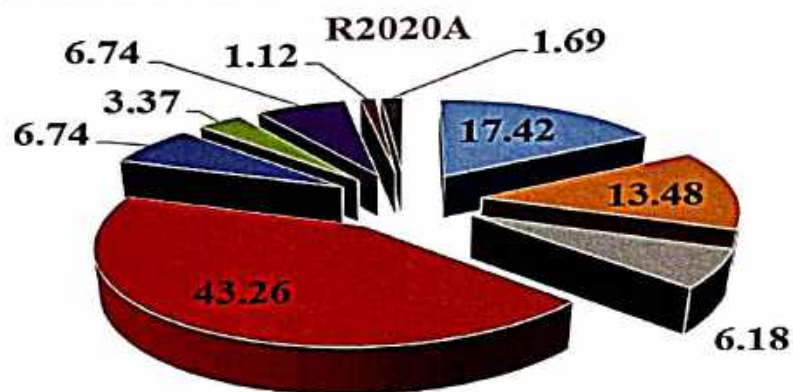
S.no	Category	SAEC R2020A credits	SAEC R2024 credits	IIT Sri city	IITD&M Kanchipuram(2020)	IIT Madras (2019)	IIT Madras (2024)
1	HSMC	9	11	24	4	27	36
2	BSC	31	27	40	24	84	64
3	PCC	77	68	36	65	120	129
4	ESC	24	19	0	11	45	35
5	EEC	17	15	9	30	72	18
6	PEC	12	18	27	12	84	60
7	OEC	6	6	14	24	0	63
TOTAL		178	165	158	170	432	405

Alignment with IIT



STRUCTURE AND COMPONENTS OF THE CURRICULUM-R2020A

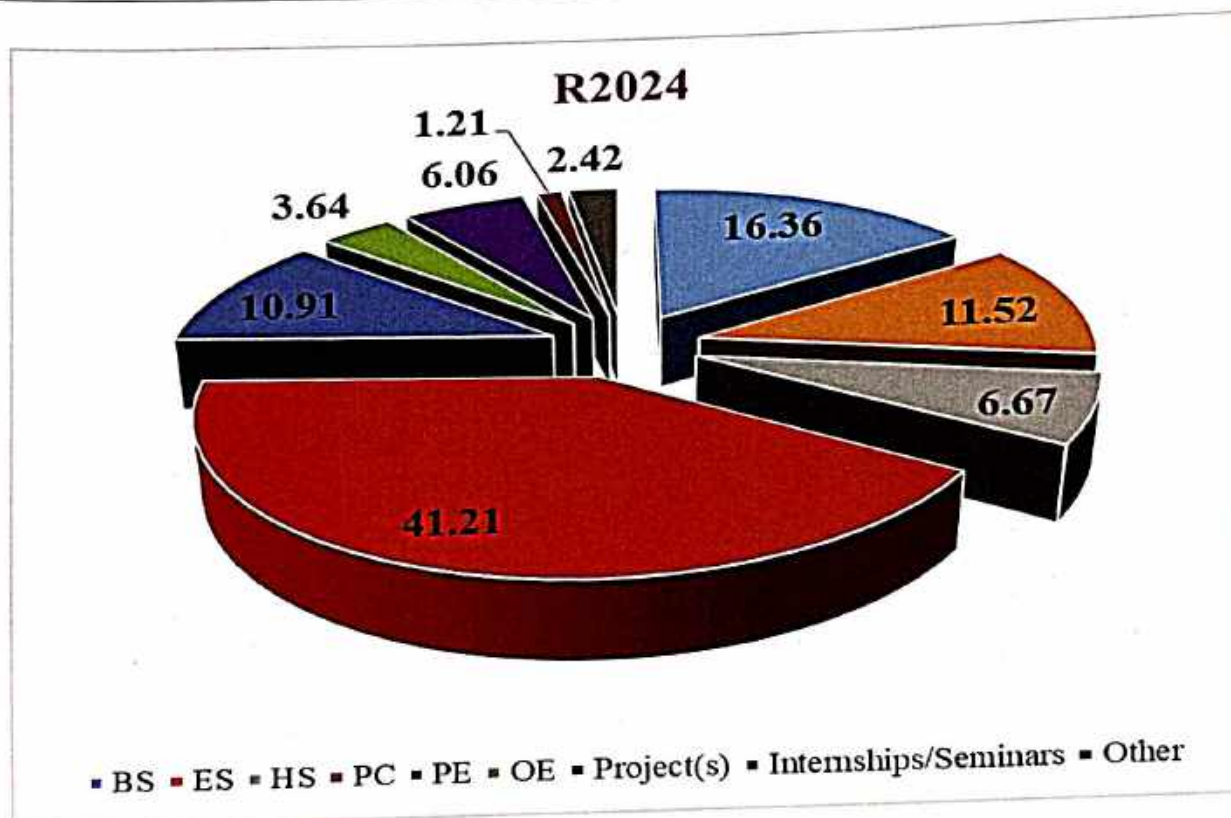
Course Component	Curriculum Content (% of total number of credits of the program)	Total number of contact hours	Total number of credits
Basic Sciences	17.42	33	31
Engineering Sciences	13.48	31	24
Humanities and Social Sciences	6.18	11	11
Program Core	43.26	100	77
Program Electives	6.74	15	12
Open Electives	3.37	6	6
Project(s)	6.74	24	12
Internships/Seminars	1.12	0	2
Any other (Please specify)	1.69	6	3
Total number of Credits			178



■ BS ■ ES ■ HS ■ PC ■ PE ■ OE ■ Project(s) ■ Internships/Seminars ■ Other

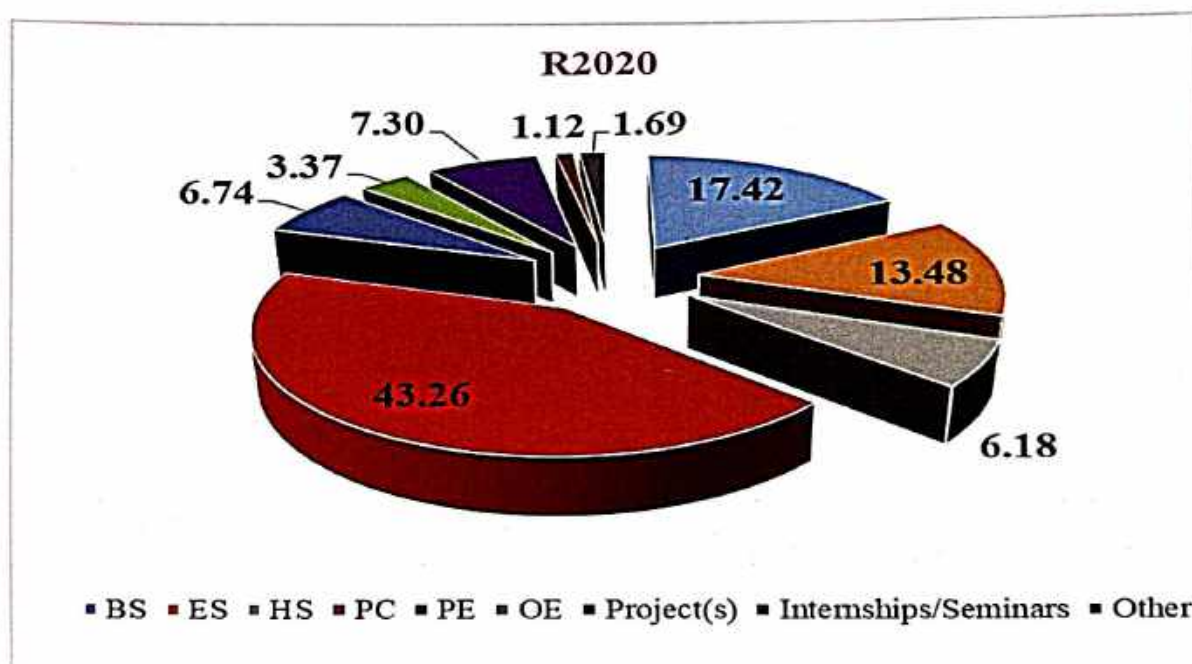
STRUCTURE AND COMPONENTS OF THE CURRICULUM-R2024

Course Component	Curriculum Content (% of total number of credits of the program)	Total number of contact hours	Total number of credits
Basic Sciences	16.36	21	27
Engineering Sciences	11.52	27	19
Humanities and Social Sciences	6.67	11	11
Program Core	41.21	85	68
Program Electives	10.91	18	18
Open Electives	3.64	6	6
Project(s)	6.06	20	10
Internships/Seminars	1.21	0	2
Any other (Please specify)	2.42	39	4
Total number of Credits			165



STRUCTURE AND COMPONENTS OF THE CURRICULUM-R2020

Course Component	Curriculum Content (% of total number of credits of the program)	Total number of contact hours	Total number of credits
Basic Sciences	17.42	33	31
Engineering Sciences	13.48	31	24
Humanities and Social Sciences	6.18	11	11
Program Core	43.26	96	77
Program Electives	6.74	12	12
Open Electives	3.37	6	6
Project(s)	7.30	26	13
Internships/Seminars	1.12	0	2
Any other (Please specify)	1.69	6	3
Total number of Credits			179



S.A ENGINEERING COLLEGE, CHENNAI – 77(Autonomous-Institute Level Research Centre, Affiliated to Anna University) DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING Parent Feedback Form

We are requesting to fill-up the form below. Your feedback is valuable to us. This will help us in improvising our services.

 arumaishiney@saec.ac.in [Switch account](#)

 Not shared 

* Indicates required question

Student Name: *

Your answer

Student Roll Number: *

Your answer

Parent Name: *

Your answer



Mobile Number *

Your answer

Parent Occupation: *

Your answer

Address *

Your answer

Email Address:

Your answer

Programme of study: *

Choose

Course of study *

Choose

Mobile Number *

Your answer

Present Occupation *

Your answer

Present Address *

Your answer

1. How do you rate the program that your ward is undergoing in terms of the load *
of the courses in different semesters?

Choose

2. How do you rate the availability of the Text and reference books in the Market? *

Choose

3. How do you rate the quality and relevance of the courses included into the *
semester?

Choose



4. How do you rate ambience of the College for effective delivery of the academic programs? *

Choose ▼

5. How do you rate the courses in terms of their relevance to the latest technologies or future technologies? *

Choose ▼

6. How do you rate the programs based on the comfort of your ward in coping with the workload? *

Choose ▼

7. How do you rate the quality of teaching in the College? *

Choose ▼

8. How do you rate the outcomes that your ward has achieved from the courses? *

Choose ▼

9. How do you rate the transparency of the evaluation system in the College? *

Choose ▼



Any suggestions for the Improvement on curriculum:

Your answer

Submit

Clear form

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Suggestions on Academics from Employer/Industry Experts

We are requesting to fill-up the form below. Your feedback is valuable to us. This will help us in improvising our services.

arumaishiney@saec.ac.in [Switch account](#)

 Not shared 

* Indicates required question

Name of Employer/Industry Expert : *

Your answer

Designation of Employer/Industry Expert

Your answer

Name of the Company/Organisation: *

Your answer

1. Rate the outcome based curriculum design in terms of graduate profile: *

Choose 

2. How well does the curriculum cater to the needs of the industry? *

Choose



3. Course content caters to enhance the skill sets of graduates *

Choose



4. The curriculum is updated and relevant to employer requirements : *

Choose



5. Rate the curriculum design and development as making students employable: *

Choose



6. The curriculum has imparted sufficient practical knowledge to the students in performing the job at work place. *

Choose



7. The curriculum has provision for training the students on usage of modern tools to solve complex engineering problems. *

Choose



8. The curriculum has a component that inculcates team work and leadership qualities among the students. *

Choose

9. The curriculum helps our students to learn and practice human values and professional ethics.

Choose

Any suggestions for the Improvement on curriculum:

Your answer

Submit

Clear form

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Google Forms



Academic Peers Feedback Form :

We are requesting to fill-up the form below. Your feedback is valuable to us. This will help us in improvising our services.

anumalshiney@saeg.ac.in [Switch account](#)

 Not shared 

* Indicates required question

Name: *

Your answer

Designation/Position *

Your answer

Organization *

Your answer

Address *

Your answer

Email Address:

Your answer



Programme of study: *

Choose

Course of study *

Choose

Mobile Number

Your answer

1. The curriculum incorporates proper sequence of courses in terms of imparting *
pre-requisite knowledge.

Choose

2. The Curriculum design emphasizes on entrepreneurship skills development. *

Choose

3. The curriculum is updated from time to time to meet the current needs of
industry/society. *

Choose

4. The curriculum provides enough number of tutorial sessions in the relevant courses.

Choose

5. The credit distribution is justified among the courses offered in the curriculum.

Choose

7. The curriculum imparts sufficient breadth of knowledge by offering wide variety of elective courses.

Choose

8. The curriculum imparts sufficient knowledge related to basic sciences and mathematics.

Choose

Any suggestions for the Improvement on curriculum:

Your answer

Submit

Clear form

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**S.A ENGINEERING COLLEGE,
CHENNAI – 77(Autonomous-Institute
Level Research Centre, Affiliated to
Anna University) DEPARTMENT OF
COMPUTER SCIENCE AND
ENGINEERING Alumni Feedback
Form**

We are requesting to fill-up the form below. Your feedback is valuable to us. This will help us in improvising our services.

* Indicates required question

1. Name: *

2. Year of Graduation: *

3. Email Address: *

4. Programme of study: *

 Dropdown

Mark only one oval.

☐ B.E.

☐ B.Tech.

5. Course of study *

Mark only one oval.

☐ CSE

☐ AIML

6. Mobile Number *

7. Present Occupation *

8. Present Address *

9. Whether undergone Higher Education *

Mark only one oval.

☐ Yes

☐ No

10. If Yes to the above question, Mention the details:

11.

*

1. The curriculum has sufficient number of relevant core subjects that laid foundation and help you in the current profession.

⊖ Dropdown

Mark only one oval.

- ☐ Strongly Agree
☐ Agree
☐ Partially Agree
☐ Disagree

12. 2. The core advanced subjects that you have learnt gave you sufficient depth of the knowledge and problem solving skills.

* ⊖ Dropdown

Mark only one oval.

- ☐ Strongly agree
☐ Agree
☐ Partially agree
☐ Disagree

13. 3. The curriculum has given you choice to select elective subjects to enhance breadth of knowledge.

* ⊖ Dropdown

Mark only one oval.

- ☐ Disagree
☐ Partially Agree
☐ Agree
☐ Strongly agree

14. 4. The syllabus taught to you in various courses reflects the current needs at your work place.

* ⌵ Dropdown

Mark only one oval.

- ☐ Disagree
☐ Partially Agree
☐ Agree
☐ Strongly Agree

15. 5. The curriculum in general is intellectually stimulating and engaging for a lifelong learning experience.

* ⌵ Dropdown

Mark only one oval.

- ☐ Disagree
☐ Partially agree
☐ Agree
☐ Strongly Agree

16. 6. The curriculum had provided you enough knowledge to get placed into a relevant company/industry.

* ⌵ Dropdown

Mark only one oval.

- ☐ Disagree
☐ Partially Agree
☐ Agree
☐ Strongly Agree

17. 7. The curriculum helped you to learn and practice human values *  Dropdown
and professional ethics.

Mark only one oval.

- ☐ Disagree
☐ Partially Agree
☐ Agree
☐ Strongly Agree

18. *
Any suggestions for the Improvement on curriculum:

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S. A. ENGINEERING COLLEGE, CHENNAI-77

(Autonomous-Institute Level Research Centre-Affiliated to Anna University)

Accredited by NBA, NAAC with "A" Grade and ISO 9001:2015 certified institution
Poonamallee - Avadi Main Road, Thiruverkadu, Chennai - 600 077.

Stakeholder Feedback & Action Taken Table (2020-2020A)

S.No	Stakeholder Group	Feedback	Impact Analysis / Gap Identification	Action Taken	Evidence / Outcome
1	Alumni	Teaching-learning can be gamified	Lack of interactive and engaging learning methods	Kahoot quizzes (2021-22), Coding Challenges (2022), Project Expo (2023)	Quiz logs, challenge reports, Expo photos
2	Alumni	Need VAC in Cybersecurity	Skills gap in security technologies	VAC on Cybersecurity & Forensics (2022), GL on Cybersecurity (2023), GL on Prompt Engineering (2024)	VAC & GL reports, feedback forms
3	Alumni	Request advanced coding practice	Placement readiness gap	Skillrack platform (2024), Ideathon (2023), Hackathon (2024)	Skillrack usage, certificates, Hackathon reports
4	Alumni	Aptitude & self-learning support	Lack of structured practice	Aptitude Buster (2023), Coursera/NPTEL (2022-24), VAC sessions (2024)	Aptitude logs, MOOC enrollment, Placements
5	Alumni	Need career guidance	Confusion about higher studies vs jobs	Career Counseling Sessions (2021-24), Alumni talks (2023)	Counseling reports, Alumni talk photos
6	Alumni	Focus on Full Stack & Cloud	Gap in modern frameworks/tools	VAC on Full Stack (2023), VAC on Cloud (2024)	VAC reports.
7	Alumni	Exposure to product development	Students lacked real product mindset	Product Development Workshop (2022), Mini Project (R2020A, 2024)	Workshop report, Mini Project reviews
8	Alumni	Expect coding contests	Low competitive coding culture	Conducted Codeathons (2022, 2023)	Contest score sheets, participation records

S.No	Stakeholder Group	Feedback	Impact Analysis / Gap Identification	Action Taken	Evidence / Outcome
9	Parents	Expectation on communication skills	Soft skills gap	Professional Communication Lab (2021), GD & Interview training (2022-24)	Lab manuals, placement training reports
10	Parents	More campus recruitment support	Concern about placement opportunities	MoUs with industries (2022-24), Industry-specific training (2023-24)	MoU copies, placement statistics
11	Parents	Encourage discipline & ethics	Student behavior gap	Ethics & Professional Values course (2021), Awareness workshops (2023)	Course syllabus, workshop reports
12	Industry	Need AI/ML exposure	Curriculum lagging behind industry	GL on AI (2022), ML Workshop (2023), AI elective (2024)	GL reports, elective syllabus
13	Industry	Cybersecurity skill demand	Placement mismatch	Industry internship in Cybersecurity (2023), Cybersecurity elective (2024)	Internship completion, syllabus copy
14	Industry	Need Data Analytics skills	Skill gap in big data tools	Data Analytics VAC (2023), elective on Data Science (2024)	VAC reports, elective enrollment
15	Industry	Demand for Design Thinking	Weak innovation mindset	Design Thinking Workshop (2022), Hackathon (2024)	Workshop report, Hackathon evaluation
16	Industry	Expect internships & exposure	Limited industry connect	Internship drive (2022-24), Industry guest lectures	Internship letters, GL attendance
17	Peer Institutions	Multidisciplinary electives	Cross-domain exposure gap	Open Electives: Telehealth Tech, Web Design, Software Testing (2022-24)	Syllabus copies, enrollment data
18	Industry	Research & publication	Limited student research output	IEEE Paper Presentation Contest (2023), Research mentoring cell (2024)	Published papers, contest reports
19	Peer Institutions	International exposure	Limited foreign resource interaction	Foreign Expert Sessions in VAC (Cybersecurity, 2022), International FDPs (2023-24)	Session reports, FDP certificates

Stakeholder Feedback & Action Taken Table (2020A–2024)

S.No	Stakeholder Group	Feedback	Gap / Impact Analysis	Action Taken (Syllabus Update in R2020A / R2024)	Evidence / Outcome
1	Alumni	Teaching-learning can be gamified	Lack of interactive and engaging learning methods	CS3303 – Design Thinking,	Activity reports
2	Alumni	Need subject specific to Cybersecurity	Skills gap in emerging security technologies	CS3702 – Ethical Hacking & Penetration Testing	Syllabus
3	Parent	Aptitude & self-learning support	Lack of structured self-learning and aptitude practice	CS36SLC – Self-Learning Course (MOOC)	Syllabus
4	Alumni	Need basic knowledge about the current software development process	Lack of knowledge on Industry exposure software development	CS3202 - Essentials and Practices of Software Development (Lab Integrated)	Syllabus
5	Alumni	Need for emerging tech exposure (AI/ML, Cybersecurity, Cloud)	Alumni suggested inclusion of modern technologies for industry readiness	CS3603 – AI & ML (Lab Integrated), CS3701 – Cryptography & Cyber Security, CS3703 – Cloud Computing (Lab Integrated)	Syllabus
6	Industry	Soft skills & professional development	Gap in employability skills	HS3401 – Universal Human Values 2, HS3505 – Professional & Career Development Lab	Syllabus
7	Parents	Focus on industry-relevant skills	Concern about job-readiness of students	CS3502 – Data Science (Lab Integrated), CS3505 – Business Analytics (PE)	Syllabus
8	Industry	Advanced coding and software engineering skills	Need for problem-solving & software development competency	CS3501 – Object-Oriented Software Engineering (Lab Integrated)	Syllabus
9	Industry	Data handling, analytics, and visualization	Gap in analytical skills	CS3502 – Data Science (Lab Integrated), CS3505 – Business Analytics (PE)	Syllabus
10	Peer Institution	Exposure to new pedagogical practices	Need to align with best practices in teaching-learning	CS3303 – Design Thinking	Syllabus

S.No	Stakeholder Group	Feedback	Gap / Impact Analysis	Action Taken (Syllabus Update in R2020A / R2024)	Evidence / Outcome
11	Industry	Exposure to UI/UX and design thinking	Need for practical design skills	CS3508 – UI/UX Design & Human-Centered Design (PE)	Syllabus
12	Industry	Exposure to digital forensics	Gap in cybersecurity practical knowledge	CS3509 – Digital and Mobile Forensics (PE)	Syllabus
13	Alumni	Exposure to mobile/web technologies	Gap in full-stack and mobile development skills	CS3605 – Mobile App Dev Lab, CS3614 – MERN Stack (PE)	Syllabus
14	Industry	Advanced IoT exposure	Industry requires hands-on projects	CS3608 – Fundamentals of IoT (PE)	Syllabus
15	Peer Institution	Balanced curriculum with electives	Concern for holistic education	Professional Electives I-IV, Open Electives I-II	Syllabus

S.No	Stakeholder Group	Feedback	Impact Analysis / Gap Identification	Action Taken	Evidence / Outcome
20	Industry	Entrepreneurship culture	Lack of startup initiatives	Incubation cell strengthened (2022), Startup Bootcamp (2023), Entrepreneurship elective (2024)	Bootcamp reports, incubation records
1	Industry	Need more exposure to Generative AI & Prompt Engineering	Lack of awareness in GenAI tools for projects and placements	Guest Lecture (GL) on Prompt Engineering and GenAI tools (July 2025)	GL report, student feedback forms
2	Alumni	Request for Placement-oriented Mock Tests	Students felt underprepared for aptitude & coding rounds	Placement Mock Test Series (Aug 2025) conducted using Skillrack & Aptitude Buster	Test score sheets, attendance records
3	Industry	Desire for Mini Hackathons within the semester	Limited practice for innovation challenges	Intra-department Hackathon (Sept 2025) conducted with problem statements from faculty and alumni	Hackathon reports, participation certificates
4	Alumni	Expect more support in Project Expo	Gap in showcasing innovative ideas	Project Expo 2025 (Aug 2025) with evaluation by industry mentors	Expo reports, evaluation rubrics
5	Industry	Suggest inclusion of Cloud & DevOps tools	Industry expects practical knowledge in CI/CD pipelines	Value Added Course (VAC) on Cloud & DevOps (July 2025) conducted with industry expert	VAC syllabus, session attendance, alumni meet minutes
6	Parents	Recommend mentoring final-year projects	Students lack industry alignment in projects	Alumni joined as Project Mentors (Aug 2025) in final year project allocation	Mentor assignment records, project reviews
7	Parents	Request for Career Guidance Sessions	Parents want clarity on higher studies & placements	Parent-Student Career Counseling Meet (Sept 2025) organized	Counseling meet report, feedback
8	Parents	More focus on Soft Skills	Employers highlight	Soft Skills Workshop (July 2025) arranged by HR trainers	Workshop report, parent meet records

S.No	Stakeholder Group	Feedback	Impact Analysis / Gap Identification	Action Taken	Evidence / Outcome
			communication gaps		
9	Industry	Need stronger exposure to AI/ML-based Startups	Students missing entrepreneurial perspective	Industry Lecture on AI Startups (Aug 2025) by startup founder	Lecture report, student attendance
10	Industry	Suggest Internship Readiness Training	Students entering internships without corporate etiquette	Pre-internship Bootcamp (June 2025) conducted	Bootcamp schedule, feedback forms
11	Industry	Demand for Cybersecurity Updates	Fast-changing cyber threats require new learning	Seminar on Latest Cybersecurity Trends (Sept 2025) delivered by industry professional	Seminar photos, report, feedback
12	Peer Institutions	Recommend Research Paper Writing Support	Gap in research publications among UG students	Research Writing Workshop (July 2025) in collaboration with peer university	Workshop handouts, student paper submissions
13	Peer Institutions	Encourage Cross-college Ideathon	Limited inter-college collaboration	Joint Ideathon with Partner Institution (Sept 2025)	MoU copy, Ideathon report, certificates

Timestamp	Designation	Orga	Email Address	Program	1. Ine	2. Ine	3. Ine	4. Ine	5. Ine	7. Ine	8. Ine
2021/03/11	Assistant	Unive	ss: gmani	Cours	curric	Curriculu	curriculu	curriculu	credit	curriculu	Any
15:46:10	G.Man Professor	Colle	ipura k@gm	Mobile	ulm design	incorp emphasize	updated	provides	n is	m imparts	suggestio
2021/10/19	i Sr.Gr.)	ge of m	ail.co veena.	study Number	Strongly	Strongly	Strongly	agree	agree	sufficient	ns for the
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24-25
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S.A. Engineering College, Chennai-77
Department of Computer Science and Engineering
Suggestions on Academics from Industry Experts/Academic Experts/Alumni/Parent

Academic Year : 2024-2025.
Date : 9/10/24
Name of the Expert : Mr. Vamsi Krishna.
Designation : Director.
Name of Organization : Beeja Academy.
Area of Specialization : Data Structures.
Years of Experience : 7 years.

S.No.	Nature of the Suggestion	Details of Suggestion
1	Feedback on Curriculum	Good.
2	Curriculum Gap	In Data Structures and OOPs, case studies can be added.

3	Student Skill Set Enhancement and Suggestion	Aptitude Skills can be enhanced through online platforms, HackerRank, Skillrack
4	Incorporate Emerging Technologies and Tools	
5	Recommend integrating practical exposure to current industrial tools and platforms to strengthen laboratory and project-based learning.	
6	Suggest inclusion of industry-oriented value-added courses, technical certifications, and collaborative projects to align student skills with emerging market demands.	


Signature of Expert


Signature of HoD

24-25
odd.

S.A. Engineering College, Chennai-77
Department of Computer Science and Engineering
Suggestions on Academics from Industry Experts/Academic Experts/Alumni/Parent ✓

Academic Year : 2024 - 2025
Date : 11.09.2024
Name of the Expert : Maria Tacintha . S
Designation : Software Developer
Name of Organization : G.E Energy
Area of Specialization : UI/UX Developer, Java
Years of Experience : 19 yrs.

S.No.	Nature of the Suggestion	Details of Suggestion
1	Feedback on Curriculum	Placement related training can be given to the students
2	Curriculum Gap	Fine. Good Curriculum

3	Student Skill Set Enhancement and Suggestion	-
4	Incorporate Emerging Technologies and Tools	Insist students to do the students projects on emerging Technologies.
5	Recommend integrating practical exposure to current industrial tools and platforms to strengthen laboratory and project-based learning.	-
6	Suggest inclusion of industry-oriented value-added courses, technical certifications, and collaborative projects to align student skills with emerging market demands.	-



Signature of Expert


Signature of HoD

24-25
odd

S.A. Engineering College, Chennai-77
Department of Computer Science and Engineering
Suggestions on Academics from Industry Experts/Academic Experts/Alumni/Parent

Academic Year : 2024-2025
 Date : 21/8/24
 Name of the Expert : Dr. P. Kalaiivan
 Designation : Associate Professor
 Name of Organization : SIMAT (Saneetha)
 Area of Specialization : Computer Science
 Years of Experience : 10 years

S.No.	Nature of the Suggestion	Details of Suggestion
1	Feedback on Curriculum	Can be improved.
2	Curriculum Gap	→ Integrate more lab component courses → Add more courses for communication improvement

3	Student Skill Set Enhancement and Suggestion	
4	Incorporate Emerging Technologies and Tools	
5	Recommend integrating practical exposure to current industrial tools and platforms to strengthen laboratory and project-based learning.	
6	Suggest inclusion of industry-oriented value-added courses, technical certifications, and collaborative projects to align student skills with emerging market demands.	Hackaton, Coding competition can be conducted to the students.

Signature of Expert

Signature of HoD

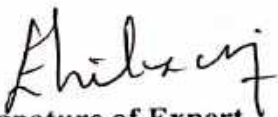
25-20
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S.A. Engineering College, Chennai-77
Department of Computer Science and Engineering
Suggestions on Academics from Industry Experts/Academic Experts/Alumni/Parent

Academic Year : 2025 - 2026 .
 Date : 01/08/25
 Name of the Expert : Dr. Ezhilavan B.
 Designation : Founder & CEO
 Name of Organization : VEI Technologies .
 Area of Specialization : Full Stack
 Years of Experience : 10 years

S.No.	Nature of the Suggestion	Details of Suggestion
1	Feedback on Curriculum	Web Technology subject has to be added.
2	Curriculum Gap	

3	Student Skill Set Enhancement and Suggestion	
4	Incorporate Emerging Technologies and Tools	MERN Stack - Electives can be added
5	Recommend integrating practical exposure to current industrial tools and platforms to strengthen laboratory and project-based learning.	
6	Suggest inclusion of industry-oriented value-added courses, technical certifications, and collaborative projects to align student skills with emerging market demands.	


Signature of Expert


Signature of HoD

S.A. Engineering College, Chennai-77
Department of Computer Science and Engineering
Suggestions on Academics from Industry Experts/Academic Experts/Alumni/Parent

Academic Year : 2023 - 2024
 Date : 5. 7. 23
 Name of the Expert : Dr. Vivek Arsholkar
 Designation : Regional Mgr.,
 Name of Organization : MathWorks Academy (ARk Info solutions)
 Area of Specialization : Machine Learning
 Years of Experience : 5 Yrs.

S.No.	Nature of the Suggestion	Details of Suggestion
1	Feedback on Curriculum	hood.
2	Curriculum Gap	further add training on python libraries & also Matlab can be given.

3	Student Skill Set Enhancement and Suggestion	-
4	Incorporate Emerging Technologies and Tools	Matlab Training / Workshop.
5	Recommend integrating practical exposure to current industrial tools and platforms to strengthen laboratory and project-based learning.	if possible include ML as a lab to get quality projects.
6	Suggest inclusion of industry-oriented value-added courses, technical certifications, and collaborative projects to align student skills with emerging market demands.	-



Signature of Expert



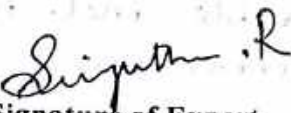
Signature of HoD

S.A. Engineering College, Chennai-77
Department of Computer Science and Engineering
Suggestions on Academics from Industry Experts/Academic Experts/Alumni/Parent

Academic Year : 23 - 24
 Date : 12.03.2024
 Name of the Expert : Ms. Sujatha . R
 Designation : Software Developer
 Name of Organization : Beeja Academy
 Area of Specialization : Full stack Developer .
 Years of Experience : 4 Years .

S.No.	Nature of the Suggestion	Details of Suggestion
1	Feedback on Curriculum	Good.
2	Curriculum Gap	Events related for Full stack Development can be Conducted & if possible incl. in the curriculum.

3	Student Skill Set Enhancement and Suggestion	encourage students to participate in hackathons.
4	Incorporate Emerging Technologies and Tools	
5	Recommend integrating practical exposure to current industrial tools and platforms to strengthen laboratory and project-based learning.	
6	Suggest inclusion of industry-oriented value-added courses, technical certifications, and collaborative projects to align student skills with emerging market demands.	


Signature of Expert


Signature of HoD

23-24 odd

S.A. Engineering College, Chennai-77
Department of Computer Science and Engineering
Suggestions on Academics from Industry Experts/Academic Experts/Alumni/Parent

Academic Year : 2023-2024.
Date : 4/8/23.
Name of the Expert : Mr. Pradeep.
Designation : Software Developer
Name of Organization : Sharpen Technologies.
Area of Specialization : Cloud computing.
Years of Experience : 10 yrs.

S.No.	Nature of the Suggestion	Details of Suggestion
1	Feedback on Curriculum	Can add more lab integrated courses.
2	Curriculum Gap	Cloud Computing Subjects can be added.

3	Student Skill Set Enhancement and Suggestion	
4	Incorporate Emerging Technologies and Tools	
5	Recommend integrating practical exposure to current industrial tools and platforms to strengthen laboratory and project-based learning.	Deployment (cloud) hands on experience can added.
6	Suggest inclusion of industry-oriented value-added courses, technical certifications, and collaborative projects to align student skills with emerging market demands.	


Signature of Expert


Signature of HoD

25-26
odd

S.A. Engineering College, Chennai-77
Department of Computer Science and Engineering
Suggestions on Academics from Industry Experts/Academic Experts/Alumni/Parent

Academic Year : 2025 - 2026
Date : 25/8/25
Name of the Expert : Dr. Sundaresan Perumal
Designation : Associate Prof
Name of Organization : University Sains Islam
Area of Specialization : Cyber Security Malaysia
Years of Experience : 12 years.

S.No.	Nature of the Suggestion	Details of Suggestion
1	Feedback on Curriculum	Good.
2	Curriculum Gap	

3	Student Skill Set Enhancement and Suggestion	
4	Incorporate Emerging Technologies and Tools	Cyber Forensics Course can be added.
5	Recommend integrating practical exposure to current industrial tools and platforms to strengthen laboratory and project-based learning.	Projects can be given in all the semesters.
6	Suggest inclusion of industry-oriented value-added courses, technical certifications, and collaborative projects to align student skills with emerging market demands.	

Sundar P. Kumar
Signature of Expert

[Signature]
Signature of HoD

23-24 even

S.A. Engineering College, Chennai-77
Department of Computer Science and Engineering
Suggestions on Academics from Industry Experts/Academic Experts/Alumni/Parent

Academic Year : 2023-2024 .
 Date : 18/3/24
 Name of the Expert : Mrs. Shobana
 Designation : Technical lead .
 Name of Organization : NIIT Limited .
 Area of Specialization : Data Science
 Years of Experience : 5 years .

S.No.	Nature of the Suggestion	Details of Suggestion
1	Feedback on Curriculum	Give more importance to lab courses .
2	Curriculum Gap	Add data Science related courses .

3	Student Skill Set Enhancement and Suggestion	Certification courses can be conducted
4	Incorporate Emerging Technologies and Tools	-
5	Recommend integrating practical exposure to current industrial tools and platforms to strengthen laboratory and project-based learning.	=
6	Suggest inclusion of industry-oriented value-added courses, technical certifications, and collaborative projects to align student skills with emerging market demands.	-


Signature of Expert


Signature of HoD

PROGRAMME ASSESSMENT COMMITTEE MEETING

Date: 22-04-2024

From : Dr.R.Geetha,HoD/CSE

Sub : Programme Assessment Committee Meeting -02

Venue : Meeting Hall

TIME :12.50 PM Dept:CSE

SEM: EVEN

Academic year : 2023-2024

The following members are hereby requested to kindly attend the Programme Assessment Meeting-II on 24-04-2024 at 12.50 in meeting Hall.

Date of Meeting

24-04-2024

Department

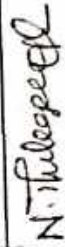
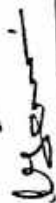
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AGENDA:

- WELCOME ADDRESS
- DISCUSSION ON PREVIOUS PAC MINUTES
- VALIDATE DAC APPROVED DEPARTMENT VISION, MISSION, AND PSO
- REPORT ON CURRICULUM GAP ANALYSIS AND MEASURES TAKEN
- REVIEW OF INNOVATIVE TEACHING-LEARNING METHODS ADOPTED BY FACULTY.
- SUMMARY OF REMEDIAL CLASSES AND MENTORING ACTIVITIES
- PLACEMENT TRAINING PROGRAMS CONDUCTED AND OUTCOMES.
- ACTIVITIES FOR SLOW LEARNERS AND ADVANCED LEARNERS.
- RESEARCH AND DEVELOPMENT

Members/Faculty Present

S/NO	Name	Designation	Signature	Reason For Absence
1.	Dr. R. GEETHA	Chairman/ HoD/Professor		
2.	Dr. M. ANURADHA	Professor		

3.	Dr.D.HEMANAND	Professor			
4.	Dr.SIVAKUMAR	Professor			
5.	Dr. A. ANTONIDOSS	Professor			
6.	Dr.D.CHITHRADEVI	Associate Professor			
7.	Dr.N.THILAGAVATHY	Associate Professor			
8.	Mr.C.BALAKRISHNAN	Associate Professor			
9.	Mr.A.MANI	Associate Professor			
10.	Mr.M.BALASUBRAMANIAN	Associate Professor			
11.	Mr.S.MUTHUKUMARASAMY	Associate Professor			
12.	Ms.V.SARASWATHI	Assistant Professor			

PREVIOUS PAC REVIEW

Sl NO	Agenda/Discussion Points	Action Plan	Action Responsibility	Target Date	Review Status
1.	REVIEW THE EXISTING VISION, MISSION, PSO, AND INITIATE REVISION	- The existing Vision, Mission, and PSOs of the department were reviewed in line with the newly reframed Institutional Vision and Mission. - The committee recommended updating them to include innovation, sustainability, and emerging technologies. - The revised drafts were prepared and forwarded to the DAC for review.	Committee Members	24.3.24	Completed
2.	IDENTIFY THE CURRICULUM GAP	PAC Committee members have suggested Guest Lectures, Workshops, Value-added courses, Seminars, etc., to bridge the curriculum gap with suitable activities for the current semester.	ECA-CCA and Club Coordinators	Continuous Process	Completed
3.	REVIEW THE MAPPING OF COs WITH POs AND PSOs	To review the previous semester's CO-PO mapping prepared by the concerned faculty members.	All Faculties	Continuous Process	Completed
4.	REVIEW OF ATTAINMENT ON POs AND PSOs	- The CO attainment of the previous semester subjects was reviewed, and a detailed gap analysis was discussed. - The PO and PSO attainment of the 2019-2023 batch was analyzed, and suitable actions were suggested for improvement. - The committee emphasized the importance of continuous monitoring to ensure the attainment levels show sustained progress.	ECA/CCA in charge and Placement Coordinators	Continuous Process	Completed

5.	CURRICULUM	- The committee reviewed the R2024 syllabus for the first and second semesters, analyzing course distribution, credit allocation, and subject relevance to ensure alignment with industry and academic requirements. - Members suggested adopting innovative teaching-learning practices such as ICT-enabled teaching, flipped classrooms, project-based learning, and the use of NPTEL/MOOCs to strengthen student engagement and learning outcomes. - The importance of strengthening lab-integrated courses was discussed, with emphasis on continuous assessment through experiments, mini-projects, and industry-relevant tools to improve practical exposure. - The elective courses for the ME program were reviewed, and it was suggested to align them with emerging domains and industry needs to enhance employability and research opportunities for students.	QMS Committee Members	End of the semester	Completed
6.	INDUSTRIAL TRAINING, INDUSTRIAL VISIT	- A six-day industrial training program was conducted for all year students in the previous semester, and will be conducted for the current semester as well. - It was suggested that industrial visits be extended to government organizations such as ISRO, DRDO, BHEL, BSNL, and NIC to expose students to large-scale projects and national initiatives. - The committee recommended organizing visits to public sector enterprises and research institutions to help students understand government-driven innovations and policies.	Placement Coordinators/Cla ss Advisors/Faculty Advisors/Faculty In-Charges	Continuous Process	Completed

7.	INTERNSHIPS AND IN-PLANT TRAININGS	- The committee discussed the details of the previous semester's internships and in-plant training that students underwent. - A total of 314 students completed regular internships across various reputed companies and organizations. - The committee appreciated the efforts taken to facilitate these opportunities and suggested exploring more industry tie-ups for diversified domains in the upcoming semesters	All faculties	Continuous Process	Completed
8.	SUMMARY OF FEEDBACK COLLECTED FROM STUDENTS, ALUMNI, INDUSTRY, AND PARENTS	- The committee reviewed the feedback from students, alumni, industry representatives, and parents from the last semester DAC meeting. - The members encouraged necessary actions to address the suggestions, focusing on enhancing teaching-learning methods, strengthening industry linkages, improving infrastructure, and supporting student skill development.	Concerned In Charges	Continuous Process	Completed
9.	DOCUMENTATION OF CONTINUOUS IMPROVEMENT PRACTICES	- Regular curriculum gap analysis, FDPs, student support initiatives, and industry engagements were reviewed and actions documented. - Continuous feedback from stakeholders and CO/PO/PSO attainment reports were analyzed to implement corrective measures for quality enhancement.	Concerned In Charges	Continuous Process	Completed

RESEARCH AND DEVELOPMENT, RESEARCH ACTIVITIES, PUBLICATIONS, AND MOUS 10.	The previous semester faculty publications and patents were discussed. Details of faculty publications:37 SCI-5 Scopus journal:12 Conference-18 Book Chapter-2 Number Of Patents Published:15 • The Committee suggested improving the patent count and working towards patent grants in the upcoming semester. • Consultancy works and funds/research projects must be initiated as suggested by the committee members.	Concerned In Charges	Continuous Process	In process
11. FACULTY TRAINING	Faculty participation in FDPs, workshops, conferences, and professional certifications during the previous semester was presented. Faculty participated in FDPs: Faculty attended NPTEL exams :18 MATLAB certification courses: It was suggested that faculty members be trained in the effective use of Learning Management Systems (LMS), virtual laboratories, simulation platforms, AR/VR-based teaching aids, and digital content development tools to enhance the quality of teaching-learning practice	All faculties	Continuous Process	In process

12.	STUDENT TECHNICAL, CO-CURRICULAR, AND OUTREACH ACTIVITIES	• Department-level technical symposiums, seminars, workshops, and hackathons that were organized during the previous semester were reviewed and planned for the upcoming semester • The committee members asked the faculty members to encourage the students to participate in inter-college competitions actively Last semester, students' participation was discussed. ➤ PAPER PRESENTATION-5 ➤ SEMINAR/WORKSHOP-29 ➤ HACKATHON-5 ➤ SYMPOSIUM-TECHNICAL EVENTS-13 ➤ DEBUGGING-5 ➤ SYMPOSIUM-NON-TECHNICAL EVENTS-4	ECA/CCA COMMITTEE MEMBERS	Completed	Completed
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Minutes of Today's Meeting

SINO	Agenda/Discussion Points	Action Plan	Action Responsibility	Target Date
1.	WELCOME ADDRESS	HoD welcomed the members present.	All Faculties	31.06.2024
2.	DISCUSSION ON PREVIOUS PAC MINUTES	The previous minutes of the meeting were discussed, and the improvements were reviewed.		
3.	VALIDATE DAC APPROVED DEPARTMENT VISION, MISSION, AND PSO	- The PAC reviewed and validated the DAC-approved Department Vision, Mission, and PSOs, confirming their alignment with industry expectations, institutional goals, and outcome-based education principles. - The committee unanimously endorsed the statements for submission to the Board of Studies (BOS) for academic approval. - As already framed, the PEO is in accordance with the new department vision and mission; the DAC members approved it and forwarded it for IQAC approval. - It was discussed that the vision, mission, and PSO should be effectively communicated to faculty, students, and stakeholders through display boards, the college website, induction programs, and orientation sessions	All Faculties	Continuous process

4.	REPORT ON CURRICULUM GAP ANALYSIS AND MEASURES TAKEN	- The following value-added courses and guest lectures were organized in the current semester. - Value Added Course on Python with MongoDB-II year - Value-added Course on Full Stack Development(Server Side) -III year - Guest Lecture on Career Guidance-III Year - Guest Lecture on Agile Technology-II Year - For the upcoming semester, the PAC committee members fixed the count of VAC, guest lectures, and seminars to be organised.	ECA/CCA Committee members	Completed
5.	REVIEW OF INNOVATIVE TEACHING-LEARNING METHODS ADOPTED BY FACULTY.	The following ICT tools were used in the current semester. - MOOC Courses - Infosys Springboard - Learning Management System is implemented for the students. Faculty were asked to prepare core course case studies and problem-based assignments for the upcoming semester	Concerned In Charges	Completed
6.	SUMMARY OF REMEDIAL CLASSES AND MENTORING ACTIVITIES	Based on the internal assessment performance, remedial classes have been conducted for the slow learners in all subjects. The faculty advisors of each class have mentored the slow learners to improve their performance.	All Faculties	In Process

7.	PLACEMENT TRAINING PROGRAMS CONDUCTED AND OUTCOMES.	• To enhance student performance in aptitude tests, technical interviews, and HIR rounds, the following training initiatives were conducted: • II Year – Aptitude Buster program focusing on quantitative, logical, and verbal reasoning skills. • III Year – Technical Training (Face Prep) emphasizing subject knowledge, coding practice, and interview readiness. • Notable increase in the number of students placed during campus recruitment drives. • Significant improvement in confidence levels, communication skills, and overall employability of students.	Placement in charge and FAs	Continuous process
8.	ACTIVITIES FOR SLOW LEARNERS AND ADVANCED LEARNERS.	For Slow Learners Remedial classes were conducted to strengthen fundamental concepts. • Meticulous counseling is to be given to the slow learner students by their Faculty advisor. • Subject staffs to call parents to discuss their performance of students for weak students. Advanced Learners are • encouraged to participate in paper presentations, hackathons, and coding contests. • Given mini-projects and case study analysis.	All faculties	Continuous Process

9.	RESEARCH AND DEVELOPMENT FACULTY RESEARCH AND DEVELOPMENT ACTIVITIES	• Faculty members are insisted to do research and development activities like publishing papers in reputed journals indexed in Scopus/SCI, increasing citation index, getting project grants, publishing patents. • Consultancy projects have been identified • Most of the faculties have published papers in SCI/SCOPUS journals or Conferences and others are also motivated to publish in SCI/Scopus indexed journals.	All faculties	Continuous Process
10.	STUDENT TECHNICAL, CO-CURRICULAR, AND OUTREACH ACTIVITIES	• Department-level technical symposiums, seminars, workshops, and hackathons were organized during the previous semester was reviewed and planned for the upcoming semester • The committee members asked the faculty members to encourage the students to participate in inter-college competitions actively • Last academic year, students' participation was discussed. • PAPER PRESENTATION-5 • PROFESSIONAL BODIES • ISTE-246 • ACM-08 • SEMINAR/WORKSHOP-29 • HACKATHON-6 • SYMPOSIUM-TECHNICAL EVENTS-20 • DEBUGGING-4 • The committee approved action plans with numerical targets for the upcoming academic year to ensure balanced academic, technical, and non-technical participation.	All faculties	Continuous Process

H/D/CSE

QMS F08C

S.A. ENGINEERING COLLEGE, CHENNAI-600 077
(Autonomous-Institute Level Research Centre-Affiliated to Anna University)
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
MINUTES OF BOARD OF STUDIES MEETING

DATE: 29-08-2025

S.A. Engineering College, Chennai conducted the meeting of Board of Studies for UG in Department of Computer Science and Engineering as per the guidelines of UGC on 29.08.2025 at 9.00 am.

The Following Members were present at the Meeting:

S.NO	NAME	DESIGNATION & ADDRESS
1.	Dr.R.Geetha (Chairperson)	Professor and HOD S.A.Engineering College
2.	Dr. S Umamaheswari (University Nominee)	Associate Professor Department of Information Technology, MIT Campus- Anna University, Chennai-6000044
3.	Dr. C Jayakumar (Subject Expert)	Associate Professor & HoD Department of Computer Science (Data Science), Rajiv Gandhi National Institute of Youth Development(RGNIYD) Ministry of Youth Affairs & Sports, Government of India Sriperumbudur Tamilnadu.- 602105.
4.	Dr V Masilamani (Subject Expert)	Professor Department of Computer Science and Engineering, Dean SA, Indian Institute of Information Technology, Design and Manufacturing Vandalur-Kelambakkam Road, Chennai- 600127.
5.	Mr. Dr R GokulaKrishnan (Representative from Industry)	Additional Director Software Technology Parks of India Chennai.
6.	Mr.B.R.SriKiranJyothi (Alumnus)	Senior Product Manager PayPal India PvtLtd,Chennai

QMS F08C

7.	DrAnuradha M	Professor
8.	Dr.Amitha T	Professor
9.	DrSivakumar S	Professor
10.	Dr.Thilagavathi N	Associate Professor
11.	Dr.Sathya Preiya V	Associate Professor
12.	Mr.Balakrishnan C	Associate Professor
13.	Mrs.Arumai Shiney S	Assistant Professor
14.	Mrs.Shobana R	Assistant Professor

- The Chairperson, Dr.R. Geetha, of the Board of Studies in the Department of Computer Science and Engineering, welcomed the members of the Board of Studies.
- The Meeting proceeded to the matters in Agenda:

S.NO	POINTS DISCUSSED & DESCRIPTION
Agenda:	
	• B.E CSE (R2024) 6th semester syllabi approval. • Discussion on new curriculum for M.E. • Discussion on Value Added Course. • Introduction of 1 credit course for foreign professor training. • Dropping of one elective course for students undergoing "Movate" company training programme.
1	Semester VI: CS3603 – Artificial Intelligence and Machine Learning (Lab Integrated): • Dr. S. Umamaheswari suggested renaming Unit 1 as "Intelligent Agent and Search Techniques" and merging it with selected topics of Unit 2. She also recommended removing the topics "History of AI – The state of the art – Risks and Benefits of AI" and dropping Dijkstra's Algorithm. • Dr. C. Jayakuma proposed reframing BFS and DFS as "Applications of BFS and DFS" instead of listing them separately. He also emphasized moving Informed Search, Greedy Best First Search, and A Algorithm* from Unit 2 into Unit 1 for better organization. • Dr. V. Masilamani suggested renaming Unit 2 as "Problem Solving and Reasoning" and adding additional topics on First Order Logic (FOL): Forward Chaining, Backward Chaining, and Resolution. • Dr. R. Gokulakrishnan highlighted that in Unit 4, the sequence would be clearer if Linear Regression and Logistic Regression are placed immediately after the general Regression topic. • Mr. B. R. Sri Kiran Jyothi recommended updating the reference section by shifting Stephen Marsland, Machine Learning – An Algorithmic Perspective, Second Edition, Chapman and Hall/CRC Machine Learning and Pattern Recognition Series, 2014 from the third reference to the second textbook. CS 3604-MOBILE APPLICATION DEVELOPMENT LABORATORY: • All members reviewed the lab structure and approved it without modifications. CS3601 COMPILER DESIGN: • Dr. V. Masilamani (Subject Expert) verified the syllabus alignment with Theory of Computation and Compiler Design and confirmed it was

acceptable.

- Dr. C. Jayakumar (Subject Expert) suggested enhancing the Suggested Activities section by explicitly including practicals on Lex and YACC using C programming language.

CS3602 – Web Technologies (Lab Integrated)

- Dr. R. Gokulakrishnan (Industry Representative) and Mr. B. R. Sri Kiran Jyothi (Alumnus) suggested that the integrated lab component be shifted to the main lab and additional exercises be included for better practical exposure.
- All members approved the theory content.

Semester 5:

CS3502-Data Science:

- “The subject Data Science is moved to the 5th semester, replacing the course Distributed and Cloud Computing. Since it is a basic subject, it is more appropriate to offer it in the 5th semester. As the 5th semester scheme was already passed in the last BoS, we are presenting it again here for clarity and approval.”
- Dr. V. Masilamani suggested renaming Unit I as “Fundamentals of Data Science”, and modifying the first topic as “Introduction to Python Libraries”.
- Dr. S. Umamaheswari proposed removing the bracketed details such as (Mean, Median, Mode) and (Range, Variance, Standard Deviation, IQR) in Unit II for conciseness.

General Discussion:

- BoS members approved the proposal for “*Movate*” to provide specialized training for students, and agreed that one elective course will be dropped for those students who opt to undergo this training.
- BoS members also approved the proposal to offer Value Added Courses for students, namely ‘RAAD – Infrastructure as Code’ for final year students, ‘Full Stack Development – Front End (MERN)’ for third year students, and ‘RAAD – DevOps & Scrum’ for second year students.
- BoS members further approved the introduction of a 1-credit Value Added Course on ‘Cyber Forensics & Incident Response: Tactical Bootcamp’ to enhance students’ practical exposure in cybersecurity.

- The Chairperson, Dr.R. Geetha, of the Board of Studies in the Department of Computer Science and Engineering, welcomed the members of the Board of Studies.
- The Meeting proceeded to the matters in Agenda:

S.NO	POINTS DISCUSSED & DESCRIPTION
Agenda: • B.E CSE (R2024) 6th semester syllabi approval. • Discussion on new curriculum for M.E.	
1	Semester VI: CS3603 – ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING (LAB INTEGRATED): • Dr. S. Umamaheswari suggested renaming Unit 1 as “Intelligent Agent and Search Techniques” and merging it with selected topics of Unit 2. She also recommended removing the topics “History of AI – The state of the art – Risks and Benefits of AI” and dropping Dijkstra’s Algorithm. • Dr. C. Jayakuma proposed reframing BFS and DFS as “Applications of BFS and DFS” instead of listing them separately. He also emphasized moving Informed Search, Greedy Best First Search, and A Algorithm* from Unit 2 into Unit 1 for better organization. • Dr. V. Masilamani suggested renaming Unit 2 as “Problem Solving and Reasoning” and adding additional topics on First Order Logic (FOL): Forward Chaining, Backward Chaining, and Resolution. • Dr. R. Gokulakrishnan highlighted that in Unit 4, the sequence would be clearer if Linear Regression and Logistic Regression are placed immediately after the general Regression topic. • Mr. B. R. Sri Kiran Jyothi recommended updating the reference section by shifting Stephen Marsland, Machine Learning – An Algorithmic Perspective, Second Edition, Chapman and Hall/CRC Machine Learning and Pattern Recognition Series, 2014 from the third reference to the second textbook. CS 3604-MOBILE APPLICATION DEVELOPMENT LABORATORY: • All members reviewed the lab structure and approved it without modifications. CS3601 COMPILER DESIGN (LAB INTEGRATED): • Dr. V. Masilamani (Subject Expert) verified the syllabus alignment with Theory of Computation and Compiler Design and confirmed it was acceptable. • Dr. C. Jayakumar (Subject Expert) suggested enhancing the laboratory section by explicitly including practical on Lex and YACC using C programming language.

CS3602 – WEB TECHNOLOGIES (LAB INTEGRATED)

- Dr. R. Gokulakrishnan (Industry Representative) and Mr. B. R. Sri Kiran Jyothi (Alumnus) suggested that the integrated lab component be shifted to the main lab and additional exercises be included for better practical exposure.
- All members approved the theory content.

PROFESSIONAL ELECTIVE AND OPEN ELECTIVE:

- All members reviewed the verticals, syllabus and approved it without modifications.

Semester 5:**CS3502-Data Science:**

- The subject Data Science is moved to the 5th semester, replacing the course Distributed and Cloud Computing. Since it is a basic subject, it is more appropriate to offer it in the 5th semester. As the 5th semester scheme was already passed in the last BoS, we are presenting it again here for clarity and approval.
- Dr. V. Masilamani suggested renaming Unit I as "Fundamentals of Data Science", and modifying the first topic as "Introduction to Python Libraries".
- Dr. S. Umamaheswari proposed removing the bracketed details such as (Mean, Median, Mode) and (Range, Variance, Standard Deviation, IQR) in Unit II for conciseness.

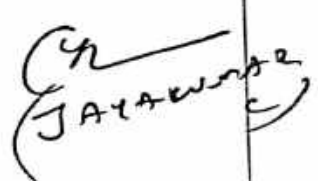
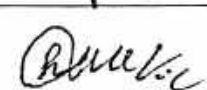
M.E revision of syllabus:

- The Board discussed the possibility of introducing a new regulation for the M.E. programme. It was observed that Anna University has been following the same regulation for the past four years, whereas our institution had planned for a revision after two years. After discussion, Dr. S. Umamaheswari suggested that, if desired, the department may proceed with framing a new regulation.

CERTIFICATE

The UG Syllabus of the Computer Science and Engineering, S.A. Engineering College (Autonomous), Chennai for 2024-2025 admission onwards, has been ratified by the Board of Studies of Computer Science and Engineering which met on 29-08-2025 at 9.00 am.

Members Present:

S.NO	NAME	DESIGNATION & ADDRESS	SIGNATURE
1.	Dr.R.Geetha (Chairperson)	Professor and HOD S.A.Engineering College	
2.	Dr. S Umamaheswari (University Nominee)	Associate Professor Department of Information Technology, MIT Campus- Anna University, Chennai-6000044	
3.	Dr. C Jayakumar (Subject Expert)	Associate Professor & HoD Department of Computer Science (Data Science), Rajiv Gandhi National Institute of Youth Development(RGNIYD) Ministry of Youth Affairs & Sports, Government of India Sriperumbudur Tamilnadu.- 602105.	
4.	Dr V Masilamani (Subject Expert)	Professor Department of Computer Science and Engineering, Dean SA, Indian Institute of Information Technology, Design and Manufacturing Vandalur-Kelambakkam Road, Chennai- 600127.	
5.	Mr. Dr R GokulaKrishnan (Representative from Industry)	Additional Director Software Technology Parks of India Chennai.	
6.	Mr.B.R.SriKiranJyothi (Alumnus)	Senior Product Manager PayPal India PvtLtd,Chennai	

7.	DrAnuradha M	Professor	M. Anuradha
8.	Dr.Amitha T	Professor	Amitha
9.	DrSivakumar S	Professor	S. S
10.	Dr.Thilagavathi N	Associate Professor	N. Thilagavathi
11.	Dr.Sathya Preiya V	Associate Professor	S. Preiya
12.	Mr.Balakrishnan C	Associate Professor	C. Balakrishnan
13.	Mrs.Arumai Shiney S	Assistant Professor	S. Arumai Shiney
14.	Mrs.Shobana R	Assistant Professor	R. Shobana



S.A. ENGINEERING COLLEGE

(Autonomous - Institute Level Research Centre)
Sponsored by Dharma Naidu Educational and Charitable Trust
Accredited by NBA, NAAC with 'A' Grade & ISO 9001 : 2015 Certified Institution
Approved by A.I.C.T.E. and Affiliated to Anna University
Poonamallee - Avadi Main Road, Thiruverkadu, Chennai - 600 077.
Phone : 2680 1999, 2680 1499, Fax : 2680 1899



**Dr.S.RAMACHANDRAN M.E., Ph.D.,
PRINCIPAL**

Date :
12.08.2025

To

Dr. C Jayakumar ,
Associate Professor & HoD
Department of Computer Science (Data Science)
Rajiv Gandhi National Institute of Youth Development(RGNIYD)
Ministry of Youth Affairs & Sports, Government of India
Sriperumbudur-602105,Tamilnadu.

Dear Sir,

Sub: Requisition to attend Board of Studies Meeting – Reg.

We implemented the Autonomous System for B.E/B.Tech/MCA/M.E (Full Time – Regular) programmes from the academic year 2020-2021. In this regard, it gives me an immense pleasure to invite you as a subject expert for the Board of Studies meeting scheduled to be conducted on 29.08.2025 from 9.30 am to 4 pm at S.A. Engineering College.

We request you to contribute your valuable suggestions in revising the existing Curriculum & Syllabus and framing the new curriculum and syllabus of 2025-2026 academic year that includes the industrial requirement and latest technology prevailing in the following departments, which can help the student community to shape themselves for their bright future.

- 1.Computer Science and Engineering
- 2.Information Technology
- 3.Artificial Intelligence & Data Science
- 4.Computer Science and Business Systems
- 5.CSE (Artificial Intelligence and Machine Learning)
6. CSE (Cyber security)
7. Master of Computer Applications

Kindly accept our invitation and confirm your presence for attending the Board of Studies Meeting.

Thank you

With Warm Regards,

PRINCIPAL

**Dr. S Ramachandran, M.E., Ph.D.,
PRINCIPAL
S.A Engineering College (Autonomous),
Poonamallee-Avadi Main Road,**





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Poonamallee - Avadi Main Road, Thiruverkadu, Chennai - 600 077.

Phone : 2680 1999, 2680 1499, Fax : 2680 1899



Date :
29.8.2025

To

Mr.B.R.SriKiranJyothi
Senior Product Manager
PayPal India PvtLtd,
Chennai.

Dear Sir,

Sub: Thanks Letter for attending Board of Studies Meeting - Reg.

I would like to express my warmth appreciation to you for giving your valuable suggestions and contributions to the Department Board of Studies Meeting that was held on 29.8.2025 in the following Departments at S.A. Engineering College.

1. Computer Science and Engineering
2. Information Technology
3. Artificial Intelligence & Data Science
4. Computer Science and Business Systems
5. CSE (Artificial Intelligence and Machine Learning)
6. CSE (Cyber security)
7. Master of Computer Applications

As an Alumnus member for framing Curriculum and Syllabus, you have been instrumental in the planning and execution of our Autonomous System implementation. Your dedication and ideas based on present needs will be incorporated in the departments to enhance the subjects and teaching methodology in the upcoming years.

Thank you once again for spending your valuable time and moments with us to provide great suggestions and updates based on current industrial scenarios.

I look forward to receive support and suggestions further more.

Thank You

With Warm Regards,

PRINCIPAL

Dr. S Ramachandran, M.E.Ph.D.,
PRINCIPAL
S.A Engineering College (Autonomous),
Poonamallee-Avadi Main Road,
Veeraragavapuram,
Thiruverkadu, Chennai-600 077.



Sub: Thanks Letter for attending Board of Studies Meeting - Eng

I would like to express my warm appreciation to you for giving your valuable suggestions and contributions to the Department Board of Studies Meeting that was held on 29.8.2023 in the following Departments at S.A. Engineering College.

1. Computer Science and Engineering
2. Information Technology
3. Artificial Intelligence & Data Science
4. Computer Science and Business Systems
5. CSI (Artificial Intelligence and Machine Learning)
6. CSI (Cyber security)
7. Master of Computer Applications

As a Subject Expert member for Training Curriculum and Syllabus, you have been instrumental in the planning and execution of our Autonomous System implementation. Your dedication and share based on present needs will be incorporated in the departments to enhance the subjects and teaching methodology in the upcoming years.

Thank you once again for spending your valuable time and moments with us to provide great suggestions and updates based on current industrial scenario.

I look forward to receiving support and suggestions for future.

With Regards,
Dr. S. Ramachandran, M.E., Ph.D.
Principal,
S.A. Engineering College,
Chennai-600 077.
Ph: 044-2680 1999.

One attached - Scanned by Gmail

Labels: Application 21, Follow up, Misc, Placement, Priority, S&EC 5/87, Students Complaints

ECA & CCA ACTIVITY PLAN

DEPARTMENT : COMPUTER SCIENCE AND ENGINEERING
 ACADEMIC YEAR : 2023-2024
 SEMESTER : Odd Semester

DATE: 28.06.2023

S.NO.	DATE / DURAT ION	NATURE OF PROGRAM(GL/IV / PROFESSIONAL BODY/CONFERE NCE)	DETAILS/TOPI C OF THE PROGRAM	DETAILS OF THE GUEST(S)	DETAILS OF THE PERTICIPANTS	VENUE	SPONSORS IF ANY
1.	2 nd week of July(5 days)	Value added course	Kotlin Programming Language	Silicon Technologies Mr.Rajendran	II CSE A, B and C	R&D LAcab	NA
2.	3 rd week of July (5 days)	Value added course	Full Stack JAVA Programming (client side)	Beeja Academy Mr.Suresh	III CSE A, B and C	LAB 7&8	NA
3.	4 th week of July (5 days)	Value added course	Cyber security	TG Cyber Labs Mr.Sriram Kesavan	IV CSE A, B and C	LAB 7&8	NA
4.	02.08.23	Guest Lecture	Guest Lecture on Advance operating System	Mr.G.JITHIN, System Engineer, TCS,Siruseri.	II CSE A,B,C	ECE / MCA Seminar Hall	NA
5.	18.08.23	Guest Lecture	Guest Lecture on “ Applications of Data Structures in Software Industry”	Mr.J.PARTHASARATHY, Director,STPL, Taramani,Chennai	II CSE A,B,C	ECE / MCA Seminar Hall	NA

S.A.ENGINEERING COLLEGE

				EC F01		
6.	10.08.23	Guest Lecture	Guest Lecture on "Recent trends in Network Security"	Palaniappan S.P Technical Architect Tech Mahindra, Chennai.	III CSE A,B,C	ECE / MCA Seminar Hall NA
7.	25.08.23	Guest Lecture	Guest Lecture on "Applying object orientation in real time environment"	Mr.Kailash, Project Manager, C-DAC, Chennai.	III CSE A,B,C	ECE / MCA Seminar Hall NA
8.	09.08.23	Guest Lecture	Guest Lecture on "Recent Trends in Cloud Computing"	Mr. Umashankar Nedunchezian CTO, I Cloud Hub Pte. Ltd. Phone: 9840395556 Email: nus@icloudhub.com	IV CSE A,B,C	ECE / MCA Seminar Hall NA
9.	31.08.23	Guest Lecture	Ethical Hacking and Cyber Security	Mr. Vinoth Senthil, Director & CTO Infysec, Chennai.	IV CSE A,B,C	ECE / MCA Seminar Hall NA
10.	30.07.20 23	One day Workshop	Cloud Computing	Mr. Mr.Kailash, Project Manager, C-DAC, Chennai.	IV Year Student	ECE / MCA Seminar Hall NA
11.	10.09.20 23	One day Workshop	Workshop on WebApp Development	Nirmal Senior Project Engineer Hitachi Chennai Contact No. 9952378549	II and III year Student	ECE / MCA Seminar Hall NA

S.A.ENGINEERING COLLEGE

EC F01

12.	06.08.20 23	Faculty Development program	Faculty Development on outcome based education	Dr.Anitha SRM University Chennai Dr.R.Geetha HoD/CSE,SAEC	For Faculty of CSE Department	ECE / MCA Seminar Hall	NA
13.	May 2024	International Conference	International conference on DeepLearning and Cyber Security	--	II and III year Student	ECE / MCA Seminar Hall	NA
14.	1 st week of Sept 23	Industrial visit	Bsnl, Chennai	Mr.Balakrishnan Senior engineer Meenambakam Chennai	II CSE A,B,C	Company	NA
15.	2 nd week of Sept 23	Industrial visit	IIITDM CHENNAI	Dr.Masilamani HoD/CSE IIITDM CHENNAI	III CSE A,B,C	Company	NA
16.	3 rd week of Sept 23	Industrial visit	CDAC CHENNAI	Mr. Mr.Kailash, Project Manager, C-DAC, Chennai.	IV CSE A,B,C	Company	NA

COORDINATOR

PRINCIPAL

S.A. ENGINEERING COLLEGE, CHENNAI-77

(An Autonomous Institution, Affiliated to Anna University)

DEPARTMENT OF CSE

ALLOCATION OF DUTIES TO FACULTY (COMMITTEE LIST)

Academic year: 2023-2024 (ODD)

Date: 04-10-2023.

Sl.No	Committee Name	Faculty In-charge	Responsibilities
1.	Poster /Pre /Post event Post event formalities	Dr.S.Sivakumar	- Preparing Pre and Post event - Website and face book Updating - Uploading the videos/photos in the department YouTube Channel after the event.
2.	Registration, Certificate design & discipline	Dr.Muthumariakshmi Ms.Lakshmi Ms. Gayathri Ms.Sreevani Ms.Dhivyabharathi	- Collecting event details and preparing poster, banner and certificate - Printing Certificate. - Collecting the participant details through coordinator. - Submitting the proposals to the juries for selection/rejection - Sending acknowledgement to the registered participants through email - Preparing the registration form and getting the signature of the participants during the start of the session - Collecting the registration form and tracking the participant's attendance - Receiving the hard copy certificate and writing names of the participants and winners - Distributing the certificates to the winners and participants - Distributing the prize - Preparing Vouchers and getting signature from the winners who receive cash prize - Maintenance of discipline till completion of event. - Gathering the students in the venue. - Ensuring students dispersal to their respective classes after event completion.
3.	Venue and Photo Committee	Ms.S.Gunanandhini Ms.R.Shobana	- Venue Arrangements and PA systems. - Software Requirements/ installation and providing assistance to the participants - Projection of banner in the venue - Ensure Mic , network availability, HDMI cable.
4.	Feedback	Ms.J.Rajalakshmi	Designing feedback form and collecting feedback hardcopy from the participants.

5.	Compering	Ms.G.Belshia Ms.S.Arumai Shiney	• Compering the event throughout the session • Giving timing instructions to the participants regularly • Taking a snapshot of all the events. • Maintaining the snapshots
6.	Hospitality	Ms.L.Sudha Ms.S.Gayathri	• Arranging refreshment for the Resource persons and participants. • Submitting the bills to budget committee.
7.	Budget, Bill Settlement	Ms.K.B.Arana	• Preparation of Budget for the event. • Distribution of funds to the required committees • Collecting the proof of expenditure • Bill settlement of the event.
8.	Quiz	Ms.Lakshmi priya	• Designing Google form with 10 MCQ related to the event. • Sharing the link and getting responses for Google form.
9.	Overall report	Mr.Balasubramanian Ms.Lakshmi priya	• Overall report submission to ECACCA committee.

HoD/CSE



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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
ORGANIZED

GUEST LECTURE

ON

CYBER SECURITY

III YEAR STUDENTS

16
APRIL
2024



044 26801999

www.saec.ac.in

saec@saec.ac.in



IBM
COMPUTER
SOCIETY



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Academy

Networking
CISCO Academy

Virtual
Labs

Microsoft



IEEE

PRO:COMM

Guest lecture on "Cyber Security"

S.A. ENGINEERING COLLEGE, CHENNAI - 77
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APPROVAL FORM

ACADEMIC YEAR: 2023-2024
SEMESTER: ODD

DATE: 17-10-2023

Department	Event Type	Event Tentative Date	Topic / Company	Whether the topic/visit relevant to semester course if yes Subject code/Name	Name of the Resource person	Company Name with address and contact no	Relevant Cos	Amount Requested (in Rs)	Remarks	HOD Sign
IT	GL	10/17/2023	UI/UX Design, Arijun Vision Tech Solutions	No	Mr. Sri Balaji	Arijun Vision Tech Solutions, Ekadachangal, Chennai.	CO6	3000	Hands on Training	<i>[Signature]</i>
CSE	GL	10/19/2023	Retech solution	CS1301 A Datastructures	Mr. M. Thanik Airwar	Retech Solution, Chennai	CO6	NIL		<i>[Signature]</i>
CSE	IV	11/6/2023	Retech solution	No	Mr. Prabhu	Retech solution, Chennai	NIL	NIL	2 nd year students	<i>[Signature]</i>
CSE	IV	11/7/2023	Retech solution	No	Mr. Prabhu	Retech solution, Chennai	NIL	NIL	2 nd year students	<i>[Signature]</i>
CSBS	IV	10/18/2023	STPI - Chennai - Software Technology Park of India	No	G. Senthil RMAG,	Software Technology Parks of India, Chennai No. 5, III Floor, Rajiv Gandhi Salai, Taramani, Chennai - 600 113	NIL	NIL		<i>[Signature]</i>
CSE	GL	11/8/2023	Greens Technology	CS1302 A- Software engineering	Mr. R. Guruprasath	Green technology, Chennai	CO6	3000		<i>[Signature]</i>

[Signature]
PRINCIPAL

[Signature]
CORRESPONDENT

[Signature]
SECRETARY



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DEPARTMENT OF CSE

organizes



GUEST LECTURE

ON

**"EXPLORING DATA STRUCTURES
ESSENTIALS"**



PARTICIPANTS

II Year Students

19
OCT
2023

M. THARIK ANWAR,
Software Developer,
Retechn Solutions, Tambaram

044 26801999

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saec@saec.ac.in



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DEPARTMENT OF CSE

organized

GUEST LECTURE HIGHLIGHTS

on "Exploring Data Structures Essentials"

PARTICIPANTS: II Year CSE Students

DATE: 19 Oct 2023



044 26801999

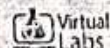
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Networking
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COMPASS





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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

ORGANIZED

INDUSTRIAL VISIT (IIIPC)

TO
TANSAM
(TAMILNADU SMART AND ADVANCED MANUFACTURING)
18 YEAR - A & B STUDENTS

06
MAR
2024



044 26801999

www.saec.ac.in

saec@saec.ac.in

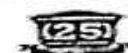


Industrial visit to TANSAM



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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

ORGANIZED

INDUSTRIAL VISIT (IIIPC)

TO
TANSAM
(TAMILNADU SMART AND ADVANCED MANUFACTURING)
18 YEAR - A & B STUDENTS

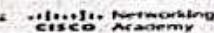
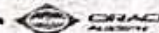
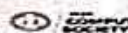
06
MAR
2024



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Industrial visit to TANSAM

21/7/25, 11:21 AM

GUEST LECTURE FEEDBACK FORM - "AGILE METHODOLOGIES"

GUEST LECTURE FEEDBACK FORM - "AGILE METHODOLOGIES"

S.A. ENGINEERING COLLEGE

An Autonomous Institution, Affiliated to Anna University, Chennai

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

GUEST LECTURE FOR "AGILE METHODOLOGIES" HELD ON 21/03/2024

Name *

Keerthana. K

Mail Id *

2216027@saec.ac.in

Register Number *

111922CS01D66

Year

☒ II

https://docs.google.com/forms/d/1PhVQ0cm-48b_SALd18d0c3ytpDLEcd14Mjnc2wplw170a-876200548/responsesACYD8NhSC00VJ_BRnQ8... 1/3

SECTION *

- ☐ A
- ☒ B
- ☐ C

1. How satisfied are you with the overall program? *

- ☒ Excellent
- ☐ Very Good
- ☐ Good
- ☐ Average

2. Participation and interaction were encouraged? *

- ☐ Strongly Agree
- ☒ Agree
- ☐ Neutral
- ☐ Disagree

2/17/25, 11:21 AM

QUEST LECTURE FEEDBACK FORM - "ACILE METHODOLOGIES"

3. On a scale of 1 to 5, (5 being the best), How would you rate the overall Program? *

- ☐ 1
☐ 2
☐ 3
☐ 4
☒ 5

4. Any other comments?

Good.....

This form was created inside of S.A. Engineering College.

Google Forms

https://docs.google.com/forms/d/1PhV00d/n-48b_BALd19du3y11OLEc14WpncZvrg/edit?ts=67b2c6b4#response=ACYDBNkSD00V1_BRnQ9... 3/3

Student feedback for Guest lecture

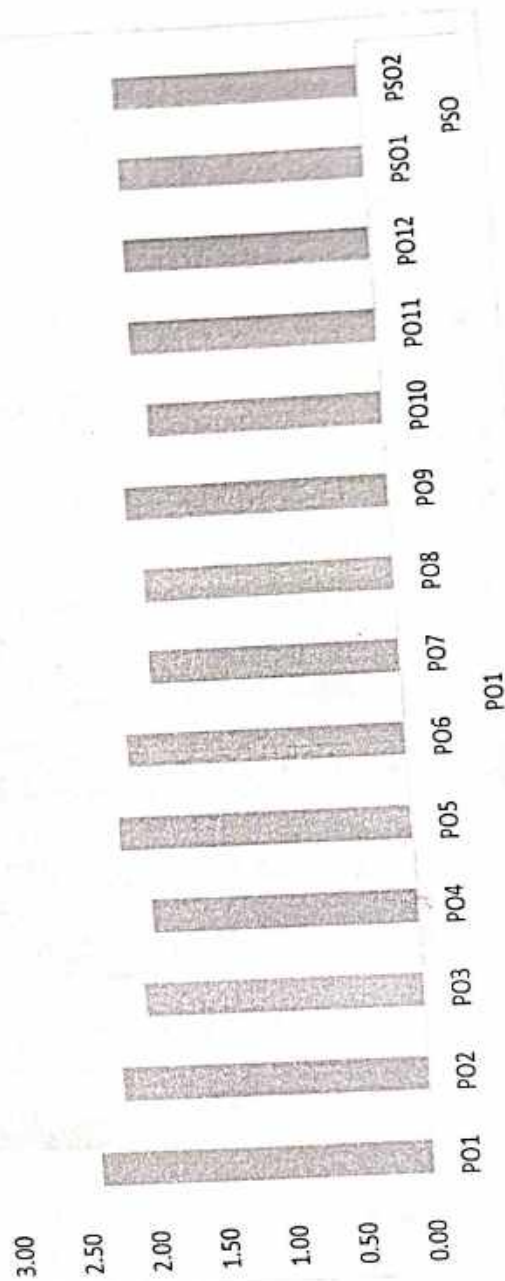
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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

2020-2024 BATCH

Direct Attainment

Assessment Methods	Tools	PO1												PSO	
		PO1												PSO	
		PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
Direct Attainment	PO & PSO through CO(70%)	2.40	2.20	2.00	1.90	2.10	2.00	1.80	1.80	1.90	1.70	1.80	1.80	1.80	1.8

Direct Attainment PO & PSO through CO(70%)



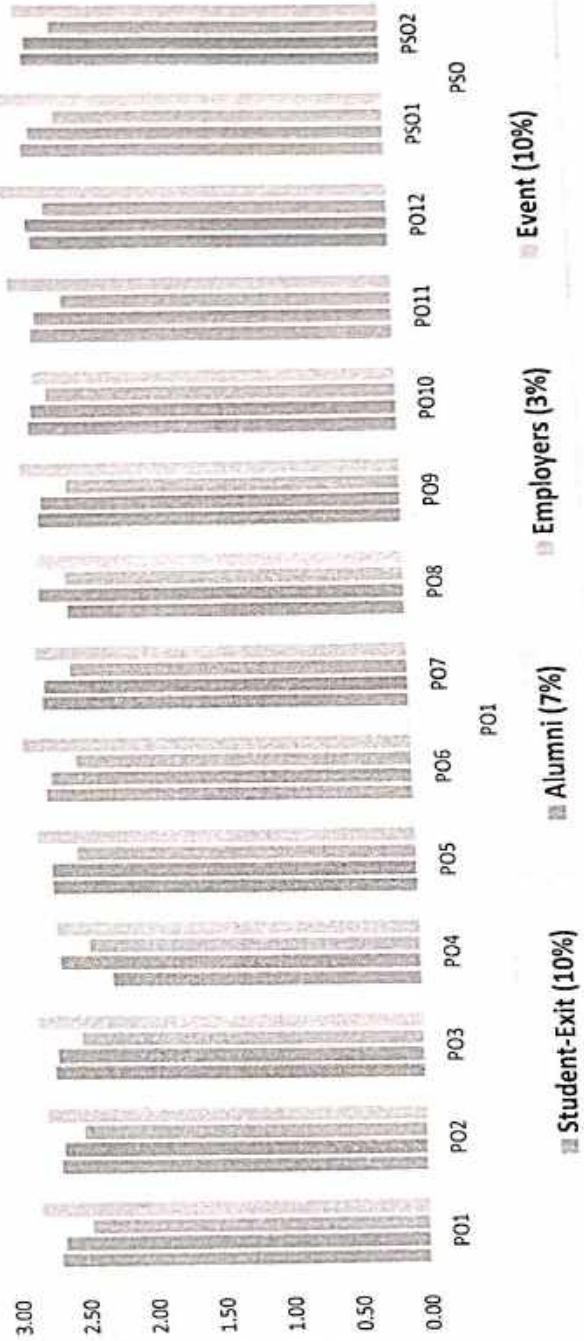
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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

2020-2024 BATCH

Indirect Attainment

Indirect Attainment															
Assessment Methods	Surveys	PO1												PSO	
		PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
Indirect Attainment	Student-Exit (10%)	2.70	2.68	2.69	2.24	2.65	2.67	2.67	2.46	2.65	2.70	2.66	2.64	2.68	2.65
	Alumni (7%)	2.67	2.65	2.67	2.62	2.65	2.63	2.65	2.67	2.63	2.68	2.63	2.67	2.63	2.63
	Employers (3%)	2.47	2.50	2.49	2.40	2.46	2.44	2.46	2.47	2.44	2.56	2.43	2.53	2.44	2.44
	Event (10%)	2.84	2.76	2.80	2.64	2.75	2.83	2.71	2.66	2.77	2.66	2.82	2.85	2.85	2.7

Indirect Attainment (30%)



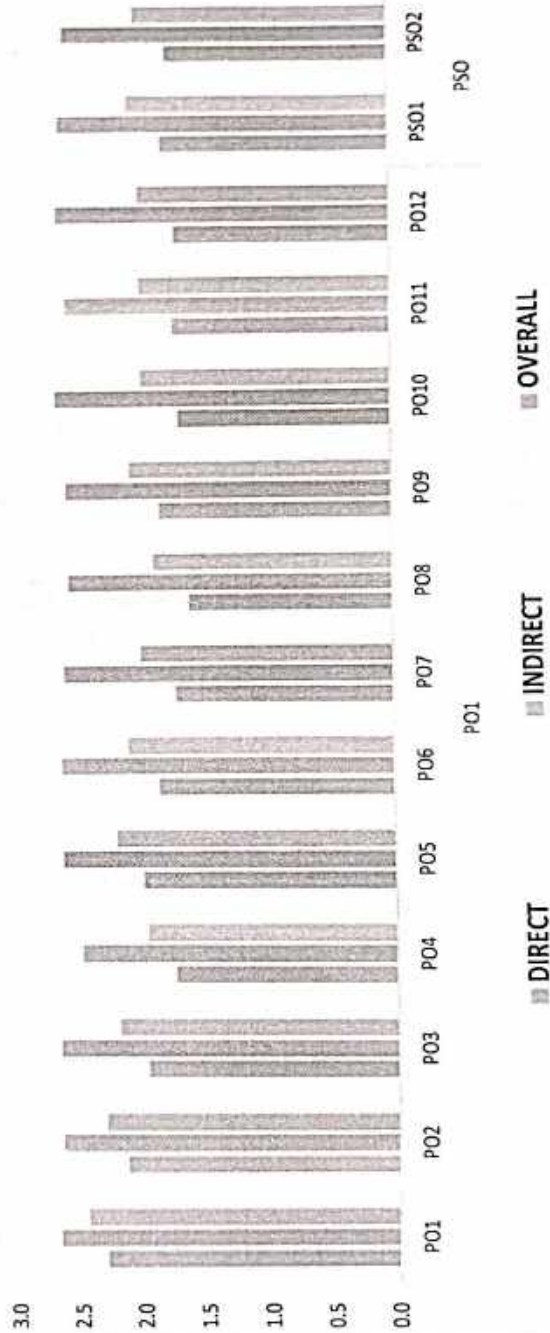
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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

2020-2024 BATCH

Direct - Indirect - Overall Attainment

Assessment Methods	PO1												PSO	
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
DIRECT	2.3	2.1	2.0	1.7	2.0	1.9	1.7	1.6	1.9	1.7	1.7	1.72	1.83	1.78
INDIRECT	2.7	2.7	2.7	2.5	2.6	2.6	2.6	2.6	2.6	2.7	2.6	2.67	2.65	2.61
OVERALL	2.5	2.3	2.2	2.0	2.2	2.1	2.0	1.9	2.1	2.0	2.0	2.01	2.1	2.04

Direct - Indirect - Overall Attainment



7.1 ACTION TAKEN BASED ON THE
RESULTS OF EVALUATION OF EACH
OF THE POs AND PSOs

CRITERION 7	Continuous Improvement	75
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1. Actions taken based on the results of evaluation of each of the COs, POs & PSOs (30)

Identify the areas of weakness in the program based on the analysis of the evaluation of COs, POs, and PSOs attainment levels. Identify and implement measures to improve POs and PSOs attainment levels for the assessment year, including curriculum interventions, pedagogical initiatives, support system improvements, etc.

POs & PSOs Attainment Levels and Actions for improvement (Batch 2019-23)

POs	Target Level	Attainment Level	Observations
PO1: Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.			
PO1	2.01	2.48	Attainment level improved against target level and hence the same method will be continued for the next year.
Action 1: Application oriented case studies given in core subjects. Action 2: Assignments on case studies given in classes. Action 3: Industry experts are invited to give sessions.			
PO2: Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.			
PO2	2.01	2.31	Attainment level improved against target level and hence same practice will be followed .
Action 1: Students are motivated to present papers at symposiums or publish in student journals. Action 2: Students are motivated to participate in technical contests (Hackathons, etc).			
PO3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.			
PO3	2.01	2.26	Attainment level improved against target level and hence the same method will be continued for the next year.

Action 1: Students are guided to do mini projects. Action 2: Partnership with industry initiated for problem statements through MoUs, internships.			
PO4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.			
PO4	2.01	2.02	Attainment level improved against target level and hence same practice will be followed
Action 1: Students are motivated to use simulation and data analysis tools (E.g. MATLAB) Action 2: Students are motivated to publish papers in reputed journals.			
PO5: Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.			
PO5	2.01	2.19	Attainment level improved against target level and hence the same method will be continued for the next year
Action 1: Students exposure to modern tools like Microsoft Azure, GitHub, AWS Cloud 9, Matlab Apache Net beans etc. will be continued. Action 2: Students are encouraged in club activities. (IoT club, Network club etc).			
PO6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice			
PO6	2.01	2.03	Attainment level improved against target level and hence same practice will be followed
Action 1: Students actively participate in Blood donation camps. Action 2: Students are associated with women empowerment programs, professional bodies like CSI, ISTE.			
PO7: Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.			
PO7	2.01	1.90	Attainment level decreased against target level
Action 1: Students to be encouraged to indulge in mini projects/projects where societal and environmental issues can be addressed. Action 2: Students are strictly instructed to switch off all electrical Equipment /Resources when not in use for all the laboratories.			
PO8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.			

PO8	2.01	2.06	Attainment level improved against target level and hence same practice will be followed
Action 1: Students learn real-world engineering failures (e.g., Bhopal Gas Tragedy, Challenger Disaster, Data Privacy Breaches) to discuss ethical lapses and lessons learned.			
PO9: Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.			
PO9	2.01	2.02	Attainment level improved against target level and hence same practice will be followed
Action 1: Students are encouraged to participate in Hackathons, Smart India Hackathon (SIH), Presentation Contests, etc. Action 2: Students are promoted to involve in departmental clubs, technical societies, and student chapters.			
P10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.			
PO10	2.01	2.04	Attainment level improved against target level and hence same practice will be followed
Action 1: Students are promoted for interactions with industry mentors through webinars, mock interviews. Action 2: Students are encouraged to participate in Hackathons, Smart India Hackathon (SIH), Paper Presentations etc			
PO11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.			
PO11	2.01	2.25	Attainment level improved against target level and hence same practice will be followed
Action 1: Students are motivated to do projects in emerging technologies and guided.			
PO12: Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.			
PO12	2.01	2.00	Attainment level decreased against target level
Action 1: Students will be motivated to do electives courses outside their specialization (open elective courses) to broaden knowledge and demonstrate adaptability.			
PSO1: To analyze, design and develop computing solutions by applying foundational concepts of Computer Science and Engineering.			

PSO1	2.01	2.03	Attainment level improved against target level and hence same practice will be followed
Action 1: Students are facilitated to do internships in reputed companies and assessed. Action 2: Students are provided with guest lectures and workshops on emerging technologies.			
PSO2: To apply software engineering principles and practices for developing quality software for scientific and business applications.			
PSO2	2.01	2.03	Attainment level improved against target level and hence same practice will be followed.
Action 1: Students learned SDLC, Agile/Waterfall methodology, design patterns, UML modelling, version control, and software testing in the subject's software engineering and object oriented analysis and design.			
PSO3: To adapt to emerging Information and Communication Technologies (ICT) to innovate ideas and solutions to existing/novel problems.			
PSO3	2.01	1.77	Attainment level decreased against target level
Action 1: More students will be encouraged to learn AWS, Azure, and do projects using , Arduino, Raspberry Pi. Action 2: More students will be motivated to join in the robotics club and IoT club and take part in the activities.			

Table 7.1.1 POs & PSOs Attainment Levels and Actions

POs & PSOs Attainment Levels and Actions for improvement (Batch 2020-24)

POs	Target Level	Attainment Level	Observations
PO1: Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.			
PO1	1.98	2.46	Attainment level improved against target level and hence the same method will be continued for the next year.
Action 1: Conduction of tutorial classes for analytical subjects will be followed by next year also. Action 2: More coaching classes will be conducted for slow learners for next academic year. Action 3: Asking viva questions relating to the basic concepts to refresh their fundamentals in laboratory sessions will be continued. Action 4: Motivation of doing internship in reputed companies will be continued. Action 5: Students will be motivated to do NPTEL and other MOOC courses, seminars, debate, concept map, quiz in respective subjects like this. Action 6: Arrangement of guest lectures will be continued on critical topics in relevant subjects to help students in identifying & analysing the real time problems.			
PO2: Problem analysis: Identify, formulate, review research literature, and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.			
PO2	1.98	2.32	Attainment level improved against target level and hence same practice will be followed .
Action 1: Relevant activities will be taken to retain the target. Action 2: Tutorial classes for analytical subjects will be conducted like this for next academic year also. Action 3: Conduction of coaching classes for slow learners will be followed. Action 4: Students will be motivated to do NPTEL and other MOOC courses, seminars, debate , concept map, quiz for next academic year also Action 5: Students will be encouraged to participate in external inter college Technical Competitions, coding contests and hackathons like this year for forthcoming academic years.			
PO3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations			
PO3	1.98	2.19	Attainment level improved against target level and hence the same method will be continued for the next year.
Action 1: Relevant activities will be taken to retain the target. Action 2: More tutorial classes for analytical subjects will be conducted like this for next academic year also. Action 3: Conduction of coaching classes for slow learners will be followed. Action 4: Conduction of Expert talks, workshop and hands on training session will be continued so that students can understand process of designing and analyzing real life software problems.			

Action 4: Students will be motivated to do NPTEL and other MOOC courses, seminars, debate , concept map, quiz for next academic year also

Action 5: Students will be encouraged to participate in external inter college Technical Competitions, coding contests and hackathons like this year for forthcoming academic years.

PO4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions

PO4	1.98	1.96	Attainment level decreased against target level
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Action 1: More awareness among students will be created for referring IEEE journal papers /Scopus to enhance their research knowledge, analysis and interpretation of data.

Action 2: Students will be motivated more to take part in national and international conferences for attaining research knowledge.

PO5: Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.

PO5	1.98	2.20	Attainment level improved against target level and hence the same method will be continued for the next year
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Action 1: Students exposure to modern tools like Microsoft Azure, GitHub, AWS Cloud 9, Matlab Apache Net beans etc. will be continued for the next academic year also.

Action 2: Students will be motivated to do NPTEL and other MOOC courses, seminars, debate, concept map, quiz for next academic year also.

Action 3: More motivation by mentors will be done for students to register for Hackathon, Capture the Flag, Paper Presentation, Technical Quiz, Webinars/Seminars conducted by other colleges regarding modern tool usage.

PO6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO6	1.92	2.14	Attainment level improved against target level and hence same practice will be followed
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Action 1: Target will be retained and will be observed for next academic year.

Action 2: More exposure of various professional engineering practices followed in the industries for students through industrial visits will be done.

Action 3: Activity based learning will be followed for next academic year also.

Action 4: Association with professional bodies like CSI, ISTE will be followed to create more awareness among the students about professional engineering practice.

Action 5: Continuation of encouragement of students will be done to carry out inter domain projects so that they would realize the importance of a project involving society, safety, health, and the legalities

PO7: Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO7	1.92	1.98	Attainment level improved against target level and hence same practice will be followed
Action 1: Continuation of encouraging students to indulge in mini projects/projects where societal and environmental issues can be addressed will be implemented. Action 2: Students are strictly instructed to switch off all electrical Equipment /Resources when not in use for all the laboratories. Action 3: On environmental context Rain water harvesting, Sewage treatment plants, Waste management procedures are employed at our college.			
PO8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.			
PO8	1.92	1.89	Attainment level decreased against target level
Action 1: Professional Ethics will be ensured by giving more importance in scrutinizing the code and project reports of Projects and Mini Projects by doing proper plagiarism check to determine the originality of the project.			
PO9: Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.			
PO9	1.92	2.09	Attainment level improved against target level and hence same practice will be followed
Action 1: Attainment level improved against target level and hence same practice will be followed Action 2: Encouragement of students to participate in various co-curricular and extra-curricular activities in other colleges/sports activities/cultural activities will be continued. Action 3: Motivating students by mentors to participate in Inter Collegiate Cultural Fest, Hackathon, Technical Quiz, Paper Presentation will be continued. Action 4: Continuation of the club activities will be done.			
PO10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.			
PO10	1.92	1.99	Attainment level improved against target level and hence same practice will be followed Attainment level improved against target level by the subject.
Action 1: Target will be retained and will be observed for next academic year. Action 2: To enhance the employability skills of the students, the training programs on the topics: mock interview, career development, higher studies motivation talk, GATE coaching classes, GATE videos by faculties ,entrepreneurship development will be continued next academic year also. Action 3: The Inter/Intra college level project exhibition will be continued for next academic year also.			
PO11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments			
PO11	1.92	2.00	Attainment level improved against target level and hence same practice will be followed

Action 1: Retention of target will be observed for next academic year. Action 2: Arrangement will be made like this year for students to visit industries on multidisciplinary environments in the next academic year to improve knowledge of the engineering principles. Action 4: Like this year students will be encouraged to write case studies as assignments to know about the recent technologies. Action 5: Students will be encouraged to prepare project proposals with the guidance of faculty for government funding agencies like this year.			
PO12: Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.			
PO12	1.92	2.01	Attainment level improved against target level and hence same practice will be followed
Action 1: Target attained will be retained Action 2: Mini Projects in curriculum will be continued which helps the students to perform better in placement interviews and higher studies Action 3: Like this year students will be motivated to register for GRE/TOEFL/GATE and other competitive examinations. Action 4: Conduction of GATE coaching classes and GATE coaching videos by faculties will be continued in next academic year also. Action 5: Students will be encouraged to publish papers in various national and International journals/conferences in next academic year also. Action 6: Students will be motivated by mentors to register and get certificates in various online courses on new technologies which would help them in continuous learning in next academic year also.			
PSO1: Apply mathematical foundations, management principles and computational concepts to design and develop efficient solutions for complex technical challenges, adhering to ethical standards and sustainable practices.			
PSO1	1.92	2.09	Attainment level improved against target level and hence same practice will be followed
Action 1: Target attained will be retained Action 2: More demonstrations, case studies, tutorial sessions and quizzes will be adapted into the teaching learning process. Action 3: More ethics related case studies will be given for the knowledge of ethical standards. Action 4: Students will be motivated to do Skill rack, Hackathon, NPTEL courses etc to know efficient solutions for complex technical solutions in next academic year like this year.			
PSO2: Analyse and design computer programs in domains such as algorithms, networking, web development, cloud computing, IoT, machine learning etc., for the development of computer-based systems with varying levels of complexity providing practical, industry-relevant solutions to meet societal and organizational needs.			
PSO2	1.92	2.04	Attainment level improved against target level and hence same practice will be followed.
Action 1: Target attained will be retained			

Action 2: More demonstrations, case studies, tutorial sessions and quizzes will be adapted into the teaching learning process

Action 3: Students will be motivated to do Skill rack NPTEL courses etc to know efficient solutions for complex technical solutions in next academic year like this year.,

Action 4: More guest lectures will be given to students in the industry related technical domains.

Table 7.1.1 POs & PSOs Attainment Levels and Actions

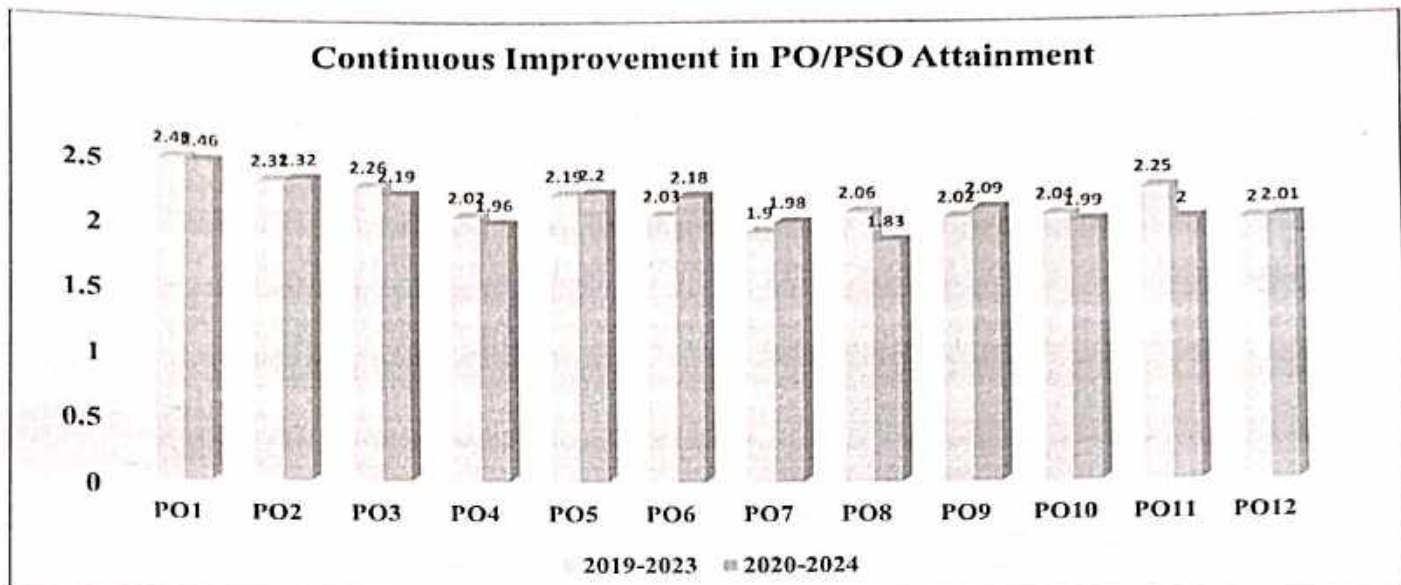


Fig.7.1a Continuous Improvement in PO/PSO attainment

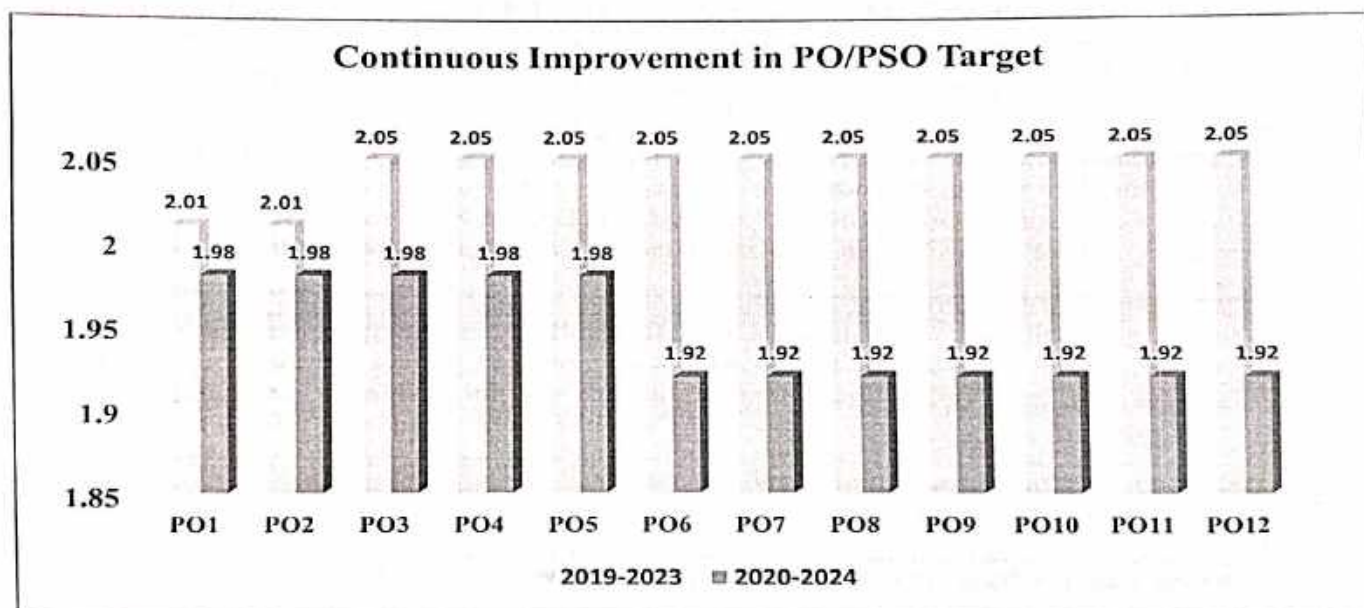


Fig.7.1b Continuous Improvement in PO/PSO target

POs & PSOs Attainment Levels and Actions for improvement (Batch 2021-25)

POs	Target Level	Attainment Level	Observations
PO1: Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.			
PO1	2.01	2.53	Attainment level improved against target level and hence the same method will be continued for the next year.
Action 1: Continue using the same effective teaching methods and assessment strategies. Action 2: Maintain regular tutorials and problem-solving sessions to strengthen core concepts. Action 3: Encourage students to apply theoretical knowledge in mini and major projects. Action 4: Conduct periodic assessments to ensure consistent understanding and improvement.			
PO2: Problem analysis: Identify, formulate, review research literature, and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.			
PO2	2.01	2.32	Attainment level improved against target level and hence same practice will be followed .
Action 1: Continue the existing teaching and learning practices that enhanced attainment. Action 2: Emphasize analytical and problem-solving skills through regular assignments. Action 3: Promote case studies and technical quizzes to improve critical thinking. Action 4: Monitor student performance continuously to maintain and further improve results.			
PO3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations			
PO3	2.01	2.21	Attainment level improved against target level and hence the same method will be continued for the next year.
Action 1: Continue implementing the current teaching-learning methods that proved effective. Action 2: Encourage students to participate in seminars and technical events to enhance Communicationskills. Action 3: Include more team-based projects to strengthen design and development abilities. Action 4: Regularly review course outcomes to ensure alignment with program objectives.			
PO4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions			

PO4	2.01	2.10	Attainment level improved against target level and hence the same method will be continued for the next year.
Action 1: Continue using the present teaching and evaluation methods that improved attainment. Action 2: Include more problem-based learning activities to enhance investigation skills. Action 3: Motivate students to use research papers and technical resources for deeper analysis. Action 4: Conduct workshops or lab sessions to strengthen experimental and analytical abilities.			
PO5: Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.			
PO5	2.01	2.31	Attainment level improved against target level and hence the same method will be continued for the next year
Action 1: Continue with the existing teaching strategies that improved the attainment level. Action 2: Encourage the use of modern tools and software in laboratory and project work. Action 3: Organize hands-on training sessions to enhance technical proficiency. Action 4: Promote the integration of simulation tools in coursework to strengthen practical understanding.			
PO6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.			
PO6	1.95	2.15	Attainment level improved against target level and hence same practice will be followed
Action 1: Continue with the present practices that contributed to improvement in attainment. Action 2: Encourage students to understand the societal and environmental impact of engineering solutions. Action 3: Include real-life case studies related to social and ethical responsibilities. Action 4: Motivate students to participate in community service and social awareness programs.			
PO7: Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.			
PO7	1.95	1.96	Attainment level improved against target level and hence same practice will be followed
Action 1: Continue the current teaching and learning approaches that led to improvement. Action 2: Create awareness among students about sustainable development and environmental practices. Action 3: Include topics and case studies related to environmental protection in coursework. Action 4: Encourage participation in eco-friendly and sustainability-related projects or			

events.			
PO8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.			
PO8	1.95	1.93	Attainment level decreased against target level
Action 1: Identify the reasons for the decline in attainment through detailed analysis of student performance. Action 2: Conduct sessions on professional ethics and responsibilities to strengthen understanding. Action 3: Introduce interactive activities like debates and case studies on ethical issues. Action 4: Include assessment components focused on ethical practices in projects and assignments. Action 5: Arrange expert lectures or workshops to improve awareness of professional and ethical behavior.			
PO9: Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.			
PO9	1.95	2.17	Attainment level improved against target level and hence same practice will be followed
Action 1: Continue the existing teaching and learning methods that improved attainment. Action 2: Encourage students to participate in team-based projects to enhance teamwork skills. Action 3: Include collaborative assignments and group discussions in the curriculum. Action 4: Monitor and provide feedback on team performance regularly. Action 5: Organize workshops or activities that strengthen leadership and interpersonal skills.			
PO10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.			
PO10	1.95	2.04	Attainment level improved against target level and hence same practice will be followed Attainment level improved against target level by the subject.
Action 1: Continue the current teaching and evaluation methods that enhanced attainment. Action 2: Promote the application of learned concepts through mini-projects and practical assignments. Action 3: Encourage active participation in seminars and technical discussions to improve communication. Action 4: Regularly assess student performance and provide constructive feedback. Action 5: Integrate real-world examples and case studies to strengthen understanding and decision-making skills.			
PO11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments			

PO11	1.95	2.18	Attainment level improved against target level and hence same practice will be followed
Action 1: Continue the existing teaching and learning methods that improved attainment. Action 2: Encourage students to engage in innovative projects to enhance problem-solving skills. Action 3: Include collaborative activities to strengthen teamwork and coordination. Action 4: Provide regular feedback on assignments and project work to guide improvement. Action 5: Organize workshops or sessions to develop critical thinking and professional skills.			
PO12: Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.			
PO12	1.95	2.09	Attainment level improved against target level and hence same practice will be followed
Action 1: Continue the current teaching and learning practices that led to improved attainment. Action 2: Encourage students to participate in research and innovation activities to enhance creativity. Action 3: Integrate problem-solving sessions and hands-on experiments in the curriculum. Action 4: Provide regular guidance and feedback on assignments and projects. Action 5: Organize seminars, workshops, and interactive sessions to strengthen analytical and professional skills.			
PSO1: Apply mathematical foundations, management principles and computational concepts to design and develop efficient solutions for complex technical challenges, adhering to ethical standards and sustainable practices.			
PSO1	1.95	2.16	Attainment level improved against target level and hence same practice will be followed
Action 1: Continue the existing teaching and learning practices that improved attainment. Action 2: Encourage students to apply software engineering principles in projects and assignments. Action 3: Conduct workshops and seminars to enhance technical knowledge and practical skills. Action 4: Provide regular feedback on student performance to ensure consistent improvement. Action 5: Integrate real-world case studies and problem-solving exercises to reinforce learning outcomes.			
PSO2: Analyse and design computer programs in domains such as algorithms, networking, web development, cloud computing, IoT, machine learning etc., for the development of computer-based systems with varying levels of complexity providing practical, industry-relevant solutions to meet societal and organizational needs.			
PSO2	1.95	2.08	Attainment level improved against target level and hence same practice will be followed.

Action 1: Maintain the current teaching methods that led to better attainment.
Action 2: Motivate students to implement ICT-based solutions in projects and assignments.
Action 3: Organize technical workshops and training sessions to enhance practical skills.
Action 4: Continuously review student performance and provide constructive feedback.
Action 5: Use real-world scenarios and problem-solving tasks to strengthen understanding and application.

PSO3	2.01	1.77	Attainment level decreased against target level
Action 1: Students will be encouraged to earn certifications (e.g., AWS Academy, Google Cloud, Microsoft Azure). Action2 : Hackathons, Ideathons, and coding challenges to be organized to encourage problem-solving Action 3: Industry experts will be invited for guest lectures and seminars on recent ICT trends and applications. Action 4: More MoUs will be signed with IT industries for internships and collaborative projects.			

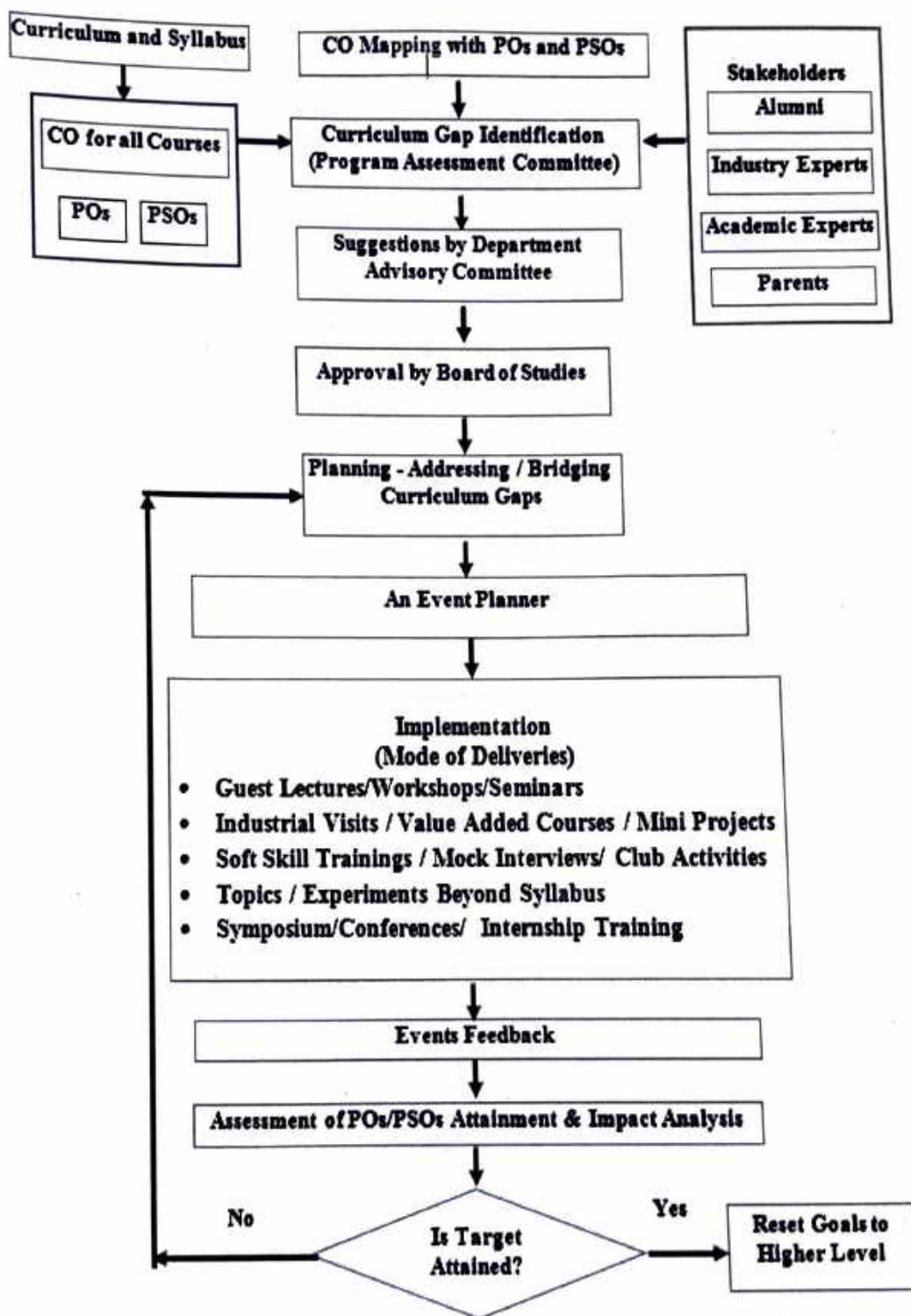


S.A ENGINEERING COLLEGE, CHENNAI - 77

(Autonomous - Institute Level Research Centre, Affiliated to Anna University)

Department of Computer Science and Engineering

Process of Curriculum Gap Identification

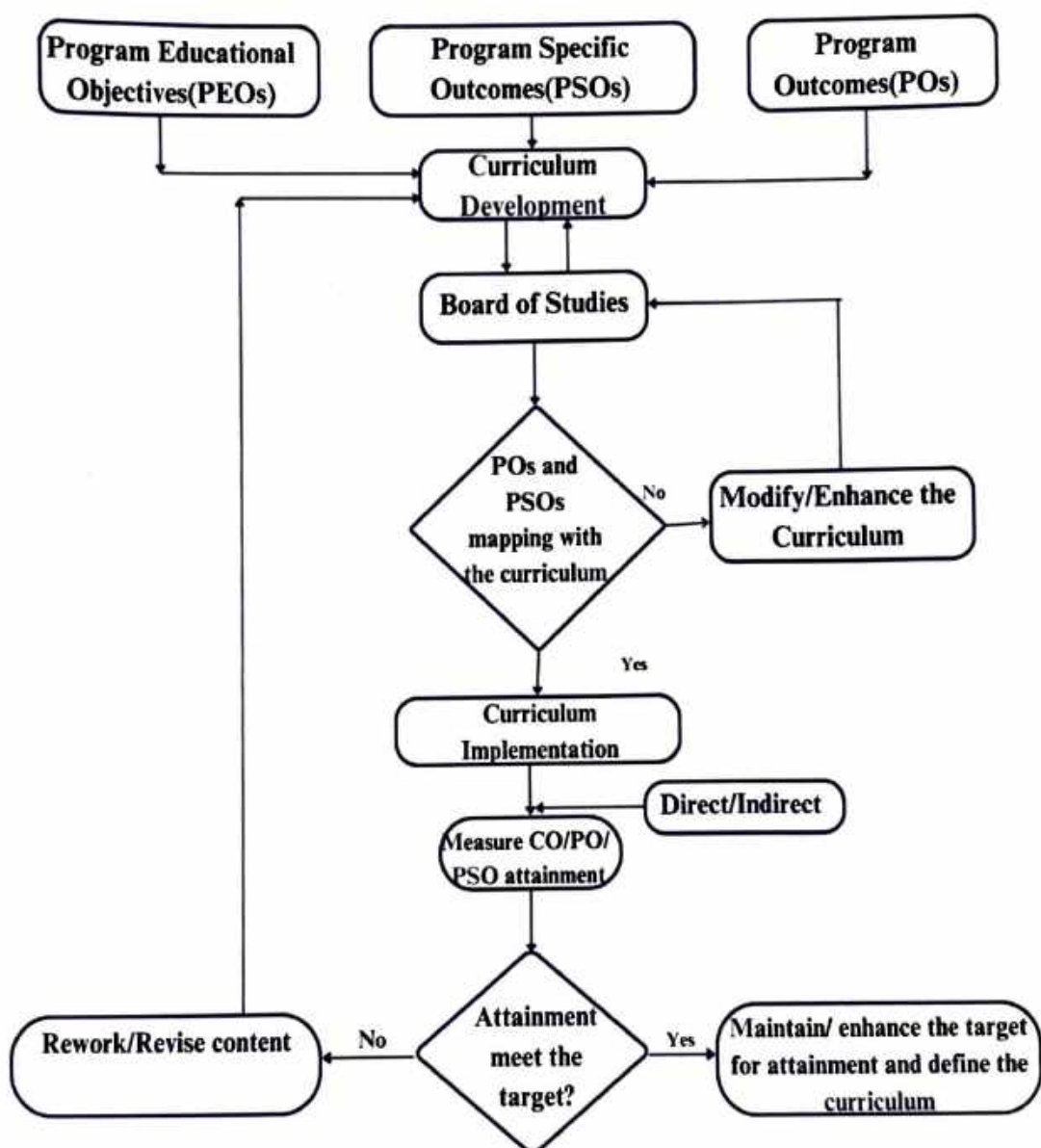




S.A ENGINEERING COLLEGE, CHENNAI - 77

(Autonomous - Institute Level Research Centre, Affiliated to Anna University)
Department of Computer Science and Engineering

PROCESS TO ENSURE THE COMPLIANCE AND ATTAINMENT OF PO'S & PSO'S



S.A ENGINEERING COLLEGE, CHENNAI – 77
(Autonomous- Institute Level Research Centre, Affiliated to Anna University)
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
REGULATION-2024

CHOICE BASED CREDIT SYSTEM
CURRICULUM & SYLLABUS



S.A. ENGINEERING COLLEGE- CHENNAI-77

(Autonomous- Institute Level Research Centre, Affiliated to Anna University)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

VISION OF THE DEPARTMENT

To pioneer in the emerging areas of Computer Science and Engineering education and research, fostering innovation, ethical values, sustainable technologies that advance the digital world and improve societal well-being.

MISSION OF THE DEPARTMENT

1. Educational Excellence:

To provide a transformative learning experience that integrates theory with practical skills, fostering creativity, critical thinking, and lifelong learning in Computer Science and Engineering.

2. Innovative Research:

To promote research and interdisciplinary collaboration, addressing contemporary challenges in technology and society.

3. Sustainable and Ethical Practices:

To facilitate the development of sustainable technological solutions with ethical practices that positively impact the environment and society.

4. Industry and Society:

To build partnerships with industry and stakeholders, enhancing the employability and entrepreneurial spirit of students, while contributing to technological and societal advancement.



S.A. ENGINEERING COLLEGE- CHENNAI-77

(Autonomous- Institute Level Research Centre, Affiliated to Anna University)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

PROGRAM EDUCATIONAL OBJECTIVES

PEO 1: Graduates shall have professional competency in the field of Computer Science and Engineering for pursuing higher education, research or as entrepreneurs.

PEO 2: Graduates shall work in a business environment with ethical standards, leadership qualities and communication necessary for engineering principles.

PEO 3: Graduates shall adapt to emerging technologies and respond to the challenges of the environment and society forever.

PROGRAM SPECIFIC OUTCOMES

PSO 1: Apply mathematical foundations, management principles and computational concepts to design and develop efficient solutions for complex technical challenges, adhering to ethical standards and sustainable practices.

PSO 2: Analyze and design computer programs in domains such as algorithms, networking, web development, cloud computing, IoT, machine learning etc., for the development of computer- based systems with varying levels of complexity providing practical, industry-relevant solutions to meet societal and organizational needs.

KNOWLEDGE AND ATTITUDE PROFILE (WK)

- WK1:** A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences.
- WK2:** Conceptually based mathematics, numerical analysis, data analysis, statistics, and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.
- WK3:** A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.
- WK4:** Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.
- WK5:** Knowledge, including efficient resource use, environmental impacts, whole- life cost, re-use of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area.
- WK6:** Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.
- WK7:** Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as the professional responsibility of an engineer to public safety and sustainable development.
- WK8:** Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues.
- WK9:** Ethics, inclusive behavior and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding and respect, and of inclusive attitudes.

PROGRAM OUTCOMES (POs)

PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.

PO2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)

PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)

PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).

PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)

PO6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).

PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)

PO8: Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

PO9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences

PO10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

PO11: Life-Long Learning: Recognize the need for, and have the preparation and ability for
i) independent and life-long learning ii) adaptability to new and emerging technologies and iii)
critical thinking in the broadest context of technological change. (WK8)

SEMESTER-I

SEMESTER-I								
S.NO	SUBJECT CODE	SUBJECT	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
THEORY								
1.	TA3101	Heritage of Tamils	HSMC	1	0	0	1	1
2.	HS3101	Professional English I	HSMC	3	0	0	3	3
3.	MA3101	Matrices and Calculus	BSC	3	1	0	4	4
4.	PH3101	Engineering Physics	BSC	3	0	0	3	3
5.	CY3101	Chemistry for Engineers	BSC	3	0	0	3	3
6.	CS3101	Programming for Problem Solving using C	ESC	3	0	0	3	3
PRACTICALS								
7.	BS3101	Physics and Chemistry Laboratory	BSC	0	0	4	4	2
8.	CS3102	Programming for Problem Solving using C Laboratory	ESC	0	0	4	4	2
9.	GE3201	Engineering Practices Laboratory	ESC	0	0	4	4	2
TOTAL				16	1	12	29	23

SEMESTER-II

S.NO	SUBJECT CODE	SUBJECT	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
THEORY								
1.	HS3201	Professional English II	HSMC	3	0	0	3	3
2.	MA3202	Discrete Mathematics	BSC	3	1	0	4	4
3.	PH3201	Physics for Information Science	BSC	3	0	0	3	3
4.	CS3201	Programming in Python	ESC	3	0	0	3	3
5.	CS3202	Essentials and Practices of Software Development (Lab Integrated)	ESC	3	0	2	5	4
6.	TA3201	Tamils and Technology	HSMC	1	0	0	1	1
7.	ME3101	Engineering Graphics	ESC	2	0	2	4	3
PRACTICALS								
8.	CS3203	Programming in Python Laboratory	ESC	0	0	4	4	2
MANDATORY COURSES								
9.	CY3201	Environmental Sciences and Sustainability	MC	1	0	0	1	1
TOTAL				19	1	8	28	24

SEMESTER III								
S. NO	SUBJECT CODE	SUBJECT	CATE GORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
THEORY				L	T	P		
1.	MA3305	Algebra and Number Theory	BSC	3	1	0	4	4
2.	CS3301	Data Structures	PCC	3	0	0	3	3
3.	CS3302	Digital Principles and Computer Organization	PCC	3	0	0	3	3
4.	CS3303	Design thinking	PCC	2	1	0	3	3
5.	IT3304	Object Oriented Programming	PCC	3	0	0	3	3
PRACTICALS								
6.	CS3304	Data Structures Laboratory	PCC	0	0	3	3	1.5
7.	IT3306	Object Oriented Programming Laboratory	PCC	0	0	3	3	1.5
EMPLOYABILITY ENHANCEMENT COURSES								
8.	HS3303	English Language Enhancement and Business Communication Laboratory	EEC	0	0	3	3	1.5
9.	CS33PT	Aptitude Skills-I	EEC	0	0	3	3	0
10.	CS33VAC	Value Added Course-I	EEC	0	0	3	3	0
TOTAL				14	2	15	31	20.5

SEMESTER IV								
S. NO.	SUBJECT CODE	SUBJECT	CATE GORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
THEORY								
1.	MA3401	Probability and Statistics	BSC	3	1	0	4	4
2.	CS3401	Computer Networks (Lab Integrated)	PCC	3	0	2	5	4
3.	IT3402	Database Management Systems	PCC	3	0	0	3	3
4.	CS3402	Design and Analysis of Algorithm (Lab Integrated)	PCC	3	0	2	5	4
5.	CS3403	Operating Systems	PCC	3	0	0	3	3
6.	HS3401	Universal Human Values - II: Understanding Harmony and Ethical Human Conduct	HSMC	2	1	0	3	3
PRACTICALS								
7.	IT3403	Database Management Systems Laboratory	PCC	0	0	3	3	1.5
8.	CS3404	Operating Systems Laboratory	PCC	0	0	3	3	1.5
EMPLOYABILITY ENHANCEMENT COURSES								
9	CS34PT	Aptitude Skills-II	EEC	0	0	3	3	0
10	CS34VAC	Value added Course-II	EEC	0	0	3	3	0
			TOTAL	17	2	16	35	24

SEMESTER V

SEMESTER V									
S. NO.	SUBJECT CODE	SUBJECT	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS	
				L	T	P			
THEORY									
1.	CS3501	Object Oriented Software Engineering (Lab Integrated)	PCC	3	0	2	5	4	
2.	CS3502	Data Science (Lab Integrated)	PCC	3	0	2	5	4	
3.	CS3503	Theory of computation	PCC	3	0	0	3	3	
4.		Professional Elective I	PEC	3	0	0	3	3	
5.		Professional Elective II	PEC	3	0	0	3	3	
6.		Open Elective I	OEC	3	0	0	3	3	
PRACTICALS									
7	HS3505	Professional and Career Development Laboratory	EEC	0	0	3	3	1.5	
EMPLOYABILITY ENHANCEMENT COURSES									
8.	CS35PT	Advanced Coding Skills-I	EEC	0	0	3	3	0	
9.	CS35VAC	Value added Course-III	EEC	0	0	3	3	0	
TOTAL				18	0	12	30	21.5	

SEMESTER VI

SEMESTER VI								
S. NO	SUBJECT CODE	SUBJECT	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
THEORY								
1.	CS3601	Compiler Design (Lab Integrated)	PCC	3	0	2	5	4
2.	CS3602	Web Technologies	PCC	3	0	0	3	3
3.	CS3603	Artificial Intelligence and Machine Learning (Lab Integrated)	PCC	3	0	2	5	4
4		Professional Elective III	PEC	3	0	0	3	3
5.		Professional Elective IV	PEC	3	0	0	3	3
6		Open Elective II	OEC	3	0	0	3	3
PRACTICALS								
7.	CS3604	Web Technologies Laboratory	PCC	0	0	3	3	1.5
8	CS3605	Mobile Application Development Laboratory	PCC	0	0	3	3	1.5
EMPLOYABILITY ENHANCEMENT COURSES								
9	CS36PT	Advanced Coding Skills-II	EEC	0	0	3	3	0
10.	CS36VAC	Value added Course-IV	EEC	0	0	3	3	0
11.	CS36SLC	Self Learning Course (MOOC Courses)	EEC					1*
TOTAL				18	0	16	34	23

SEMESTER VII

SEMESTER VII								
S. NO	SUBJECT CODE	SUBJECT	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
THEORY								
1.	CS3701	Cryptography and Cyber Security (Lab Integrated)	PCC	3	0	2	5	4
2.	CS3702	Ethical Hacking and Penetration Testing	PCC	3	0	0	3	3
3.	CS3703	Cloud Computing (Lab Integrated)	PCC	3	0	2	5	4
4.		Climate Change and Sustainability	MC	2	0	0	2	0
5.		Professional Elective V	PEC	3	0	0	3	3
6.		Professional Elective VI	PEC	3	0	0	3	3
EMPLOYABILITY ENHANCEMENT COURSES								
7.	CS3704	Project work I	EEC	0	0	4	4	2
8.	CS37INT	Summer internship	EEC	0	0	0	0	2
9.	CS37VAC	Value added Course-V	EEC	0	0	3	3	0
10.	CS37PLT	Advanced coding skills-V	EEC	0	0	3	3	0
MANDATORY COURSES								
11.	MC3701	Indian Constitution	MC	2	0	0	2	0
TOTAL				19	0	14	33	21

SEMESTER VIII

SEMESTER VIII								
S. NO	SUBJECT CODE	SUBJECT	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
PRACTICALS								
1.	CS3801	Project Work II	EEC	0	0	16	16	8
TOTAL				0	0	16	16	8

PROFESSIONAL ELECTIVE I

PROFESSIONAL ELECTIVE I									
S.NO.	COURSE CODE	COURSE TITLE	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS	
THEORY				L	T	P			
1.	CS3504			Information Security and Management	PE	3			0
2.	CS3505	Business Analytics	PE	3	0	0	3	3	
3.	CS3506	Multimedia and animation	PE	3	0	0	3	3	
4.	CS3507	3D Printing and Design	PE	3	0	0	3	3	
5.	CS3508	UI / UX Design and Human Centered Design	PE	3	0	0	3	3	

PROFESSIONAL ELECTIVE II

S. NO	COURSE CODE	COURSE TITLE	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
THEORY								
1.	CS3509	Digital and Mobile Forensics	PE	3	0	0	3	3
2.	CS3510	Social Network Analysis	PE	3	0	0	3	3
3.	CS3511	Augmented Reality and Virtual Reality	PE	3	0	0	3	3
4.	CS3512	Robotic Process Automation	PE	3	0	0	3	3
5.	CS3513	Software Testing and Automation	PE	3	0	0	3	3

PROFESSIONAL ELECTIVE III

S. NO.	COURSE CODE	COURSE TITLE	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
THEORY								
1.	CS3606	Recommender System	PE	3	0	0	3	3
2.	CS3607	Video Creation and Editing	PE	3	0	0	3	3
3.	CS3608	Fundamentals of Internet of Things	PE	3	0	0	3	3
4.	CS3609	Web Application Security	PE	3	0	0	3	3
5.	CS3610	Principles of Management	PE	3	0	0	3	3

PROFESSIONAL ELECTIVE IV								
S. NO.	COURSE CODE	COURSE TITLE	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
THEORY								
1.	CS3611	Network Firewall	PE	3	0	0	3	3
2.	CS3612	Text and Speech Analytics	PE	3	0	0	3	3
3.	CS3613	Visual Effects	PE	3	0	0	3	3
4.	CS3614	MERN Stack	PE	3	0	0	3	3
5.	CS3615	Digital Marketing	PE	3	0	0	3	3

OPEN ELECTIVE								
S. NO.	COURSE CODE	COURSE TITLE	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
THEORY								
1.	OCS3501	Relational Database Systems	OE	3	0	0	3	3
2.	OCS3502	Creative Web Design	OE	3	0	0	3	3
3.	OCS3601	Principles of Operating Systems	OE	3	0	0	3	3
4.	OCS3602	Software Engineering Methodologies	OE	3	0	0	3	3

TOTAL CREDITS:165

SUMMARY

SL. NO	SUBJECT AREA	CREDITS PER SEMESTER								CREDITS TOTAL	PERCENTAGE
		I	II	III	IV	V	VI	VII	VIII		
1.	BSC	12	7	4	4					27	16.36
2.	ESC	7	12							19	11.5
3.	PCC			15	17	11	14	11		68	41.21
4.	PEC					6	6	6		18	10.91
5.	OEC					3	3			6	3.64
6.	EEC			1.5		1.5		4	8	15	9.09
7.	MC		1							1	0.00
8.	HSMC	4	4		3					11	7.27
TOTAL		23	24	20.5	24	21.5	23	21	8	165	-

S.A ENGINEERING COLLEGE, CHENNAI – 77
(An Autonomous Institution Affiliated to Anna University)
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
REGULATION-2020A
CHOICE BASED CREDIT SYSTEM
CURRICULUM AND SYLLABUS

PROGRAM EDUCATIONAL OBJECTIVES (PEOs):

1. Graduates shall have professional competency in the field of Computer Science and Engineering for pursuing higher education, research or as entrepreneurs.
2. Graduates shall work in a business environment with ethical standards, leadership qualities and communication necessary for engineering principles.
3. Graduates shall adapt to emerging technologies and respond to the challenges of the environment and society forever.

PROGRAM OUTCOMES (POs):

Engineering Graduates will be able to:

1. **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.
2. **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5. **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
6. **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and

need for sustainable development.

8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9. **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10. **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11. **Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
12. **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs):

- Apply mathematical foundations, management principles and computational concepts to design and develop efficient solutions for complex technical challenges, adhering to ethical standards and sustainable practices.
- Analyze and design computer programs in domains such as algorithms, networking, web development, cloud computing, IoT, machine learning etc., for the development of computer-based systems with varying levels of complexity providing practical, industry-relevant solutions to meet societal and organizational needs.

Mapping of POs/PSOs to PEOs

Contribution

1: Reasonable

2: Significant

3: Strong

MAPPING OF PROGRAM EDUCATIONAL OBJECTIVES WITH PROGRAM OUTCOMES

A broad relation between the program objective and outcomes is given in the following table:

Program Educational Objectives	Program Outcomes											
	1	2	3	4	5	6	7	8	9	10	11	12
1	3	3	3	3	3	2	2	3	3	3	2	3
2	2	2	2	3	2	2	1	3	3	3	3	2
3	3	3	3	3	3	1	3	1	2	3	2	3

MAPPING OF PROGRAM SPECIFIC OBJECTIVES WITH PROGRAM OUTCOMES

A broad relation between the program specific objective and outcomes is given in the following table:

Program Specific Objectives	Program Outcomes											
	1	2	3	4	5	6	7	8	9	10	11	12
1	3	3	3	2	3	3	3	3	2	2	3	3
2	3	3	3	2	3	2	2	2	2	2	2	3

MAPPING OF COURSE OUTCOMES WITH PROGRAM OUTCOMES

A broad relation between the Course Outcomes and Program Outcomes is given in the following table

	COURSE TITLE	PROGRAMME OUTCOME(PO)											
		1	2	3	4	5	6	7	8	9	10	11	12
YEAR-I	SEMESTER-I	Technical English	-	-	-	-	-	√	√	-	√	√	√
		Calculus and its applications	√	√	√	√	-	-	-	-	-	-	-
		Applied Physics	√	√	-	-	-	-	-	-	-	-	-
		Engineering Chemistry	√	√	√	-	-	-	-	-	-	-	-
		Problem Solving and Python Programming	√	√	√	√	√	-	-	-	-	√	√
		Engineering Graphics	√	√	√	√	√	√	√	√	√	√	√
		Tamil Marabu/ Heritage of Tamils											
		Physics and Chemistry Laboratory	√	√	√	√	-	-	-	-	-	-	-
		Problem Solving and Python Programming Laboratory	√	√	√	√	-	-	-	-	-	-	√
		Indian Constitution											
	SEMESTER-II	English for Communication	-	-	-	-	√	√	-	-	√	√	-
		Complex Variables and Transforms	√	√	√	-	-	-	-	-	-	-	-
		Material Science	√	√	-	-	-	-	√	-	-	-	-
		Basic Electrical, Electronics and Measurement Engineering	√	√	√	-	-	-	-	-	-	-	-
		Programming in C	√	√	√	√	-	-	-	-	-	√	√
		Fundamentals of Computing	√	√	√	-	-	-	-	-	-	-	√
		Tamil Thozhil Nutpam / Tamils and Technology											
		Programming in C Laboratory	√	√	√	√	-	-	-	-	-	-	-
		Engineering Practice Laboratory	√	√	√	√	√	√	-	√	√	√	-
		Fundamentals of Computing Laboratory	√	√	√	√	√	-	-	√	√	√	√
		Environmental Science and Engineering	√	√	√	-	-	-	√	√	√	√	-

YEAR-II	SEMESTER-III	COURSE TITLE	PROGRAMME OUTCOME(PO)											
			1	2	3	4	5	6	7	8	9	10	11	12
		Discrete Mathematics	√	√	√	√	-	-	-	-	-	-	-	-
		Object Oriented Programming	√	√	√	√	√	-	-	-	√	-	-	√
		Data Structures	√	√	√	-	-	-	-	-	-	-	-	√
		Digital Principles and System Design	√	√	√	√	√	√	√	√	-	-	√	√
		Software Engineering	√	√	√	-	√	-	-	√	√	√	√	√
		Object Oriented Programming Laboratory	√	√	√	-	√	-	-	-	√	√	-	√
		Data Structures Laboratory	√	√	√	-	-	-	-	-	√	√	-	√
		Interpersonal Skills Laboratory	√	√	√	√	√	√	√	√	√	√	√	√
	SEMESTER-IV	Probability and Statistics	√	√	√	√	-	-	-	-	-	-	-	√
		Computer Architecture	√	√	√	-	-	-	-	-	-	-	-	√
		Database Management Systems	√	√	√	-	-	-	-	-	-	-	-	√
		Design and Analysis of Algorithms	√	√	√	-	-	-	-	-	-	-	-	√
		Universal Human Values	√	-	-	-	-	√	√	-	√	√	-	√
		Object Oriented Analysis and Design	√	√	√	√	√	-	-	-	√	√	√	√
		Database Management Systems Laboratory	√	√	√	-	√	-	-	-	√	√	-	√
		Design and Analysis of Algorithms Laboratory	√	√	√	-	-	-	-	-	√	√	-	√
		Employability and Soft Skills Laboratory	√	√	√	√	√	√	√	√	√	√	√	√

YEAR -III	SEMESTER-V	COURSE TITLE	PROGRAMME OUTCOME(PO)											
			1	2	3	4	5	6	7	8	9	10	11	12
		Algebra and Number Theory	√	√	√	-	-	-	-	-	-	-	-	-
		Computer Networks	√	√	√	-	-	-	-	-	-	-	-	√
		Operating Systems	√	√	√	-	-	-	-	-	-	-	-	√
		Theory of Computation	√	√	√	-	-	-	-	-	-	-	-	√
		Embedded System	√	√	√	√	√	-	-	-	√	√	√	√
		Open Elective I												
		Networks Laboratory	√	√	√	-	-	-	-	-	√	√	-	√
		Operating Systems Laboratory	√	√	√	-	-	-	-	-	-	-	-	-

YEAR -III	SEMESTER-VI	Internet Programming	√	√	√	-	√	-	√	-	√	√	√	√
		Compiler Design	√	√	√	-	√	-	-	-	-	-	√	√
		Artificial Intelligence	√	√	√	√	-	-	-	-	-	√	√	√
		Cryptography and Network Security	√	√	√	√	√	-	-	-	√	√	-	√
		Professional Elective I												
		Internet Programming Laboratory	√	√	√	√	√	-	√	-	√	√	√	√
		Mini Project	√	√	√	√	√	-	√	-	√	√	√	√
		Security Laboratory	√	√	√	√	√	-	-	-	-	-	-	√
		Professional Communication Laboratory	-	-	-	-	√	√	√	√	√	√	√	√

PROGRAMME OUTCOME(PO)														
YEAR -IV	SEMESTER -VII	COURSE TITLE	1	2	3	4	5	6	7	8	9	10	11	12
		Distributed Systems and cloud computing	√	√	√	-		-	√	√	√	√	-	√
		Machine Learning	√	√	√		√	-	-	-	-	-	-	-
		Principles of Management	√		√	-	-	-	√	√	√	√	√	√
		Open Elective II												
		Professional Elective II												
		Project work – I	√	√	√	√	√	√	√	√	√	√	√	√
		Cloud Computing Laboratory	√	√	√	√	√	-	√	-	√	√	√	√
		Summer internship	√	√	√	√	√	√	√	√	√	√	√	√
		Machine Learning Laboratory	√	√	√	√	√	-	-	-	-	√	-	√
	SEMESTER -VIII	Professional Elective III												
		Professional Elective IV												
		Project Work-II	√	√	√	√	√	√	√	√	√	√	√	√

PROFESSIONAL ELECTIVES

SEM	COURSE TITLE	PROGRAMME OUTCOME(PO)											
		1	2	3	4	5	6	7	8	9	10	11	12
VI	Cyber Security	√	√	√	√	√	√	√	√	√	√	√	√
	Software Testing	√	√	√	√	-	-	√	-	-	-	-	-
	Agile Methodologies	√	√	√								√	
	Graph Theory and Applications	√	√	√									
	Data warehousing and Data mining	√	√	√									
VII	Service Oriented Architecture	√	√	√									
	Total Quality Management	√	√	√								√	
	Multi-core Architectures and Programming	√	√	√									
	Human Computer Interaction	√	√	√	√	√	√	√	√	√	√	√	√
	C# and .Net Programming	√	√	√	√	√	√	-	-	-	-	-	√
	Wireless Adhoc and Sensor Networks	√	√	√									
	Advanced Topics on Databases	√	√	√									
	Full Stack Web Development	√	√	√	-	√	√	√	-	√	√	√	√
	Human Rights	√	√	√									
VIII	Blockchain Technologies	√	√	√	√	√	√	√	√	-	-	-	√
	Neural Networks and Deep Learning	√	√	√	√	√	√			√	√	√	√
	Professional Ethics in Engineering						√	√	√	√	√		√
	Big Data Analytics	√	√	√		√				√	√		
	Principles of Programming Languages	√	√	√									√
	Security and Privacy in Cloud	√	√	√	√	√				√	√	√	√
	Game Development	√	√	√	√	√	√						
	Multimedia and Animation	√	√	√	√	√	√			√	√	√	√
VIII	UI and UX Design	√	√	√	√	-	-	-	-	-	-	-	-
	Green Computing	√	√	√	√	√	√	√	-	√	√	√	√
	Mobile and Pervasive Computing	√	√	√									

SL.N		Software Defined Networks	✓	✓	✓									
THEO		Digital marketing	✓	✓	✓	-	✓	-	-	-	✓	-	-	✓
1.		Knowledge Engineering	✓	✓	✓									✓
2.		Ethical Hacking	✓	✓	✓	✓	✓	✓		✓				✓

SEMESTER-I

SL.NO.	SUBJECT CODE	SUBJECT	CATEGORY	CONTACT PERIODS	L	T	P	C
THEORY								
1.	HS1101A	Technical English	HS	3	3	0	0	3
2.	MA1101A	Calculus and its applications	BS	4	3	1	0	4
3.	PH1101A	Applied Physics	BS	3	3	0	0	3
4.	CY1101A	Engineering Chemistry	BS	3	3	0	0	3
5.	CS1101A	Problem Solving and Python Programming	ES	3	3	0	0	3
6.	ME1101A	Engineering Graphics	ES	4	2	0	2	3
7.	TA1101A	Tamils and Technology/ Tamilar Thozhil Nutpam /	HSMC	1	1	0	0	1
PRACTICALS								
8.	BS1101A	Physics and Chemistry Laboratory	BS	4	0	0	4	2
9.	CS1102A	Problem Solving and Python Programming Laboratory	ES	4	0	0	4	2
MANDATORY COURSE								
10	CI1101A	Indian Constitution	MC	2	2	0	0	0
TOTAL				31	20	1	10	24

SEMESTER-II

SL.NO.	SUBJECT CODE	SUBJECT	CATEGORY	CONTACT PERIODS	L	T	P	C
THEORY								
1.	HS1201A	English for Communication	HS	3	3	0	0	3
2.	MA1201A	Complex Variables and Transforms	BS	4	3	1	0	4
3.	PH1201A	Materials Science	BS	3	3	0	0	3
4.	EE1202A	Basic Electrical, Electronics and Measurement Engineering	ES	3	3	0	0	3
5.	CS1201A	Programming in C	ES	3	3	0	0	3
6.	CS1202A	Fundamentals of Computing	PC	3	3	0	0	3
7.	TA1201A	Tamizhar Marabu	HSMC	1	1	0	0	1
PRACTICALS								
8.	CS1203A	Programming in C Laboratory	ES	4	0	0	4	2
9.	GE1201A	Engineering Practices Laboratory	ES	4	0	0	4	2
10.	CS1204A	Fundamentals of Computing Laboratory	PC	2	0	0	2	1
MANDATORY COURSE								
11.	CY1201A	Environmental Science and Engineering	MC	2	2	0	0	0
TOTAL				32	21	1	10	25

SEMESTER -III								
SL.NO	SUBJECT CODE	SUBJECT	CATEGORY	CONTACT PERIODS	L	T	P	C
THEORY								
1.	MA1303A	Discrete Mathematics	BS	4	3	1	0	4
2.	IT1301A	Object Oriented Programming	PC	3	3	0	0	3
3.	CS1301A	Data Structures	PC	3	3	0	0	3
4.	EC1306A	Digital Principles and System Design	ES	3	3	0	0	3
5.	CS1302A	Software Engineering	PC	3	3	0	0	3
PRACTICALS								
6.	IT1302A	Object Oriented Programming Laboratory	PC	4	0	0	4	2
7.	CS1303A	Data Structures Laboratory	PC	4	0	0	4	2
8.	HS1301A	Interpersonal Skills Laboratory	EEC	2	0	0	2	1
TOTAL				26	15	1	10	21

SEMESTER-IV								
SL.NO	SUBJECT CODE	SUBJECT	CATEGORY	CONTACT PERIODS	L	T	P	C
THEORY								
1.	MA1403A	Probability and Statistics	BS	4	3	1	0	4
2.	CS1401A	Computer Architecture	PC	3	3	0	0	3
3.	IT1401A	Database Management Systems	PC	3	3	0	0	3
4.	CS1402A	Design and Analysis of Algorithms	PC	3	3	0	0	3
5.	HV1401A	Universal Human Values	ES	3	2	1	0	3
6.	CS1403A	Object Oriented Analysis and Design	PC	5	3	0	2	4
PRACTICALS								
7.	IT1402A	Database Management Systems Laboratory	PC	4	0	0	4	2
8.	CS1404A	Design and Analysis of Algorithms Laboratory	PC	4	0	0	4	2
9.	HS1401A	Employability And Soft Skills Laboratory	EEC	2	0	0	2	1
TOTAL				31	17	2	12	25

SEMESTER-V								
SL.NO	SUBJECT CODE	SUBJECT	CATEGORY	CONTACT PERIODS	L	T	P	C
THEORY								
1.	MA1501A	Algebra and Number Theory	BS	4	3	1	0	4
2.	IT1501A	Computer Networks	PC	3	3	0	0	3
3.	CS1501A	Operating Systems	PC	3	3	0	0	3
4.	CS1502A	Theory of Computation	PC	3	3	0	0	3
5.	EC1515A	Embedded System	PC	3	3	0	0	3
6.		Open Elective-I	OE	3	3	0	0	3
PRACTICALS								
7.	IT1503A	Networks Laboratory	PC	4	0	0	4	2
8.	CS1503A	Operating Systems Laboratory	PC	4	0	0	4	2
9.	CS15VAC	Value Added Course	EEC	1	0	0	0	1*
TOTAL				27	18	1	8	23

SEMESTER -VI

SL.NO.	SUBJECT CODE	SUBJECT	CATEGORY	CONTACT PERIODS	L	T	P	C
THEORY								
1.	CS1601A	Internet Programming	PC	3	3	0	0	3
2.	CS1602A	Compiler Design	PC	5	3	0	2	4
3.	CS1603A	Artificial Intelligence	PC	3	3	0	0	3
4.	CS1604A	Cryptography and Network Security	PC	3	3	0	0	3
5.		Professional Elective-I	PE	3	3	0	0	3
PRACTICALS								
6.	CS1605A	Internet Programming Laboratory	PC	4	0	0	4	2
7.	CS1606A	Security Laboratory	PC	4	0	0	4	2
8.	CS1612A	Mini Project	EEC	4	0	0	4	2
9.	HS1601A	Professional Communication Laboratory	EEC	2	0	0	2	1
TOTAL				31	15	0	16	23

SEMESTER-VII

SL.NO.	SUBJECT CODE	SUBJECT	CATEGORY	CONTACT PERIODS	L	T	P	C
THEORY								
1.	IT1701A	Distributed Systems and Cloud Computing	PC	3	3	0	0	3
2.	CS1701A	Machine Learning	PC	3	3	0	0	3
3.	MG1701A	Principles of Management	HS	3	3	0	0	3
4.		Open Elective-II	OE	3	3	0	0	3
5.		Professional Elective-II	PE	3	3	0	0	3
PRACTICALS								
6.	CS1702A	Machine Learning Laboratory	PC	4	0	0	4	2
7.	IT1702A	Cloud Computing Laboratory	PC	4	0	0	4	2
8.	CS1703A	Summer Internship	EEC	0	0	0	0	2
9.	CS1714A	Project work – I	EEC	6	0	0	6	3
TOTAL				29	15	0	14	24

SEMESTER-VIII

SL.NO.	SUBJECT CODE	SUBJECT	CATEGORY	CONTACT PERIODS	L	T	P	C
THEORY								
1.		Professional Elective-III	PE	3	3	0	0	3
2.		Professional Elective-IV	PE	3	3	0	0	3
PRACTICALS								
3.	CS1801A	Project Work-II	EEC	14	0	0	14	7
TOTAL				20	6	0	14	13

TOTAL NO. OF CREDITS: 178

HUMANITIES AND SOCIAL SCIENCES (HS)

SL.NO.	SUBJECT CODE	SUBJECT	CATEGORY	CONTACT PERIODS	L	T	P	C
1.	HS1101A	Technical English	HS	3	3	0	0	3
2.	HS1201A	English for Communication	HS	3	3	0	0	3
3.	MG1701A	Principles of Management	HS	3	3	0	0	3

HUMANITIES, SOCIAL SCIENCES AND MANAGEMENT COURSES (HSMC)

SL.NO.	SUBJECT CODE	SUBJECT	CATEGORY	CONTACT PERIODS	L	T	P	C
1.	TA1101A	Tamil Marabu/ Heritage of Tamils	HSMC	1	1	0	0	1
2.	TA1201A	Tamil Thozhil Nutpam / Tamils and Technology	HSMC	1	1	0	0	1

BASIC SCIENCES (BS)

SL.NO.	SUBJECT CODE	SUBJECT	CATEGORY	CONTACT PERIODS	L	T	P	C
1.	MA1101A	Calculus and its applications	BS	4	3	1	0	4
2.	PH1101A	Applied Physics	BS	3	3	0	0	3
3.	CY1101A	Engineering Chemistry	BS	3	3	0	0	3
4.	BS1101A	Physics and Chemistry Laboratory	BS	4	0	0	4	2
5.	MA1201A	Complex Variables and Transforms	BS	4	3	1	0	4
6.	PH1201A	Material Science	BS	3	3	0	0	3
7.	MA1303A	Discrete Mathematics	BS	4	3	1	0	4
8.	MA1403A	Probability and Statistics	BS	4	3	1	0	4
9.	MA1501A	Algebra and Number Theory	BS	4	3	1	0	4

ENGINEERING SCIENCES (ES)

SL.NO.	SUBJECT CODE	SUBJECT	CATEGORY	CONTACT PERIODS	L	T	P	C
1.	CS1101A	Problem Solving and Python Programming	ES	3	3	0	0	3
2.	ME1101A	Engineering Graphics	ES	4	2	0	2	3
3.	CS1102A	Problem Solving and Python Programming Laboratory	ES	4	0	0	4	2
4.	EE1202A	Basic Electrical, Electronics and Measurement Engineering	ES	3	3	0	0	3
5.	CS1201A	Programming in C	ES	3	3	0	0	3
6.	CS1203A	Programming in C Laboratory	ES	4	0	0	4	2
7.	GE1201A	Engineering Practices Laboratory	ES	4	0	0	4	2
8.	EC1306A	Digital Principles and System Design	ES	3	3	0	0	3
9.	HV1401A	Universal Human Values	ES	3	2	1	0	3

EMPLOYABILITY ENHANCEMENT COURSES (EEC)

SL.NO.	SUBJECT CODE	SUBJECT	CATEGORY	CONTACT PERIODS	L	T	P	C
1.	HS1301A	Interpersonal Skills Laboratory	EEC	2	0	0	2	1
2.	HS1401A	Employability And Soft Skills Laboratory	EEC	2	0	0	2	1
3.	HS1601A	Professional Communication Laboratory	EEC	2	0	0	2	1
4.	CS1703A	Summer Internship	EEC	0	0	0	0	2
5.	CS1612A	Mini Project	EEC	4	0	0	4	2
6.	CS1704A	Project work – I	EEC	6	0	0	6	3
7.	CS1801A	Project Work-II	EEC	14	0	0	14	7

PROFESSIONAL CORE (PC)

SL.NO.	SUBJECT CODE	SUBJECT	CATEGORY	CONTACT PERIODS	L	T	P	C
1.	CS1202A	Fundamentals of Computing	PC	3	3	0	0	3
2.	CS1204A	Fundamentals of Computing Laboratory	PC	2	0	0	2	1
3.	IT1301A	Object Oriented Programming	PC	3	3	0	0	3
4.	CS1301A	Data Structures	PC	3	3	0	0	3
5.	CS1302A	Software Engineering	PC	4	0	0	4	2
6.	IT1302A	Object Oriented Programming Laboratory	PC	4	0	0	4	2
7.	CS1303A	Data Structure Laboratory	PC	3	3	0	0	3
8.	CS1401A	Computer Architecture	PC	3	3	0	0	3
9.	IT1401A	Database Management Systems	PC	3	3	0	0	3
10.	CS1402A	Design and Analysis of Algorithms	PC	5	3	0	2	4
11.	CS1403A	Object Oriented Analysis and Design	PC	4	0	0	4	2
12.	IT1402A	Database Management Systems Laboratory	PC	4	0	0	4	2
13.	CS1404A	Design and Analysis of Algorithms Laboratory	PC	3	3	0	0	3
14.	IT1501A	Computer Networks	PC	3	3	0	0	3
15.	CS1501A	Operating Systems	PC	3	3	0	0	3
16.	CS1502A	Theory of Computation	PC	3	3	0	0	3
17.	EC1515A	Embedded System	PC	4	0	0	4	2
18.	IT1503A	Networks Laboratory	PC	4	0	0	4	2
19.	CS1503A	Operating Systems Laboratory	PC	3	3	0	0	3
20.	CS1601A	Internet Programming	PC	5	3	0	2	4
21.	CS1602A	Compiler Design	PC	3	3	0	0	3
22.	CS1603A	Artificial Intelligence	PC	3	3	0	0	3
23.	CS1604A	Cryptography and Network Security	PC	3	3	0	0	3
24.	CS1605A	Internet Programming Laboratory	PC	4	0	0	4	2

25.	CS1606A	Security Laboratory	PC	4	0	0	4	2
26.	IT1701A	Distributed Systems and Cloud Computing	PC	3	3	0	0	3
27.	CS1701A	Machine Learning	PC	3	3	0	0	3
28.	CS1702A	Machine Learning Laboratory	PC	4	0	0	4	2
29.	IT1702A	Cloud Computing Laboratory	PC	4	0	0	4	2

PROFESSIONAL ELECTIVES

ELECTIVE-I

SL.NO.	SUBJECT CODE	SUBJECT	CATEGORY	CONTACT PERIODS	L	T	P	C
THEORY								
1.	CS1607A	Cyber Security	PE	3	3	0	0	3
2.	CS1608A	Software Testing	PE	3	3	0	0	3
3.	CS1609A	Agile Methodologies	PE	3	3	0	0	3
4.	CS1610A	Graph Theory and Applications	PE	3	3	0	0	3
5.	CS1611A	Data warehousing and Data mining	PE	3	3	0	0	3

ELECTIVE-II

SL.NO.	SUBJECT CODE	SUBJECT	CATEGORY	CONTACT PERIODS	L	T	P	C
THEORY								
1.	CS1704A	Service Oriented Architecture	PE	3	3	0	0	3
2.	CS1705A	Total Quality Management	PE	3	3	0	0	3
3.	CS1706A	Multi-core Architectures and Programming	PE	3	3	0	0	3
4.	CS1707A	Human Computer Interaction	PE	3	3	0	0	3
5.	CS1708A	C# and .Net Programming	PE	3	3	0	0	3
6.	CS1709A	Wireless Adhoc and Sensor Networks	PE	3	3	0	0	3
7.	CS1710A	Advanced Topics on Databases	PE	3	3	0	0	3
8.	CS1711A	Full Stack Web Development	PE	3	3	0	0	3
9.	CS1712A	Human Rights	PE	3	3	0	0	3
10.	CS1713A	Blockchain Technologies	PE	3	3	0	0	3

ELECTIVE-III

SL.NO.	SUBJECT CODE	SUBJECT	CATEGORY	CONTACT PERIODS	L	T	P	C
THEORY								
1.	CS1802A	Neural Networks and Deep Learning	PE	3	3	0	0	3
2.	GE1801A	Professional Ethics in Engineering	PE	3	3	0	0	3
3.	CS1803A	Big Data Analytics	PE	3	3	0	0	3
4.	CS1804A	Principles of Programming Languages	PE	3	3	0	0	3
5.	CS1805A	Security and Privacy in Cloud	PE	3	3	0	0	3
6.	CS1806A	Game Development	PE	3	3	0	0	3
7.	CS1807A	Multimedia and Animation	PE	3	3	0	0	3

ELECTIVE-IV

SL.NO.	SUBJECT CODE	SUBJECT	CATEGORY	CONTACT PERIODS	L	T	P	C
THEORY								
1.	CS1808A	UI and UX Design	PE	3	3	0	0	3
2.	CS1809A	Green Computing	PE	3	3	0	0	3
3.	CS1810A	Mobile and Pervasive Computing	PE	3	3	0	0	3
4.	CS1811A	Software Defined Networks	PE	3	3	0	0	3
5.	CS1812A	Digital Marketing	PE	3	3	0	0	3
6.	CS1813A	Knowledge Engineering	PE	3	3	0	0	3
7.	CS1814A	Ethical Hacking	PE	3	3	0	0	3

OPEN ELECTIVE OFFERED TO OTHER DEPARTMENT
OPEN ELECTIVE I
SEMESTER V

SL.NO.	SUBJECT CODE	SUBJECT	CATEGORY	CONTACT PERIODS	L	T	P	C
THEORY								
1.	OCS501A	Software Engineering	OE	3	3	0	0	3
2.	OCS502A	Data Structures and Algorithms	OE	3	3	0	0	3
3.	OCS503A	Telehealth Technology	OE	3	3	0	0	3

OPEN ELECTIVE II
SEMESTER VII

SL.NO.	SUBJECT CODE	SUBJECT	CATEGORY	CONTACT PERIODS	L	T	P	C
THEORY								
1.	OCS701A	Software Testing	OE	3	3	0	0	3
2.	OCS702A	Web Design and management	OE	3	3	0	0	3

SUMMARY

SL.NO.	Subject Area	Credits per semester								Credits Total	Percentage
		I	II	III	IV	V	VI	VII	VIII		
1.	HS	3	3					3		9	5.1
2.	BS	12	7	4	4	4				31	17.4
3.	ES	8	10	3	3					24	13.5
4.	PC		4	13	17	16	17	10		77	43.3
5.	PE						3	3	6	12	6.7
6.	OE					3		3		6	3.4
7.	EEC			1	1		3	5	7	17	9.6
8.	MC	0	0							0	0.0
9.	HSMC	1	1							2	1.1
TOTAL		24	25	21	25	23	23	24	13	178	-